DAET TOURIST INFORMATION, ENGAGEMENT, AND MANAGEMENT SYSTEM WITH REAL-TIME UPDATES

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CHAPTER 1

INTRODUCTION

This chapter presents the foundational framework of the study. It describes the system being developed, the conditions that made it necessary, the organization it serves, the problems it addresses, the objectives it pursues, and the boundaries within which it operates.

Project Context

The Daet Tourist Information, Engagement, and Management System with Real-Time Updates is a web-based Progressive Web Application (PWA) that serves as a single official online portal for tourism in the Municipality of Daet, Camarines Norte. It replaces the current patchwork of social media posts, static pages, and unofficial blogs with one source of verified tourism information that is maintained directly by the Daet Municipal Tourism Office.

The project builds three connected components. The first is a public tourism portal where visitors can browse and search attractions, accommodations, restaurants, local businesses, and events, view photos, maps, operating hours, entrance fees, and contact details, and get directions to each place. The second is a real-time alert and engagement layer that delivers official advisories (weather, sea and

beach conditions, road closures, power and water interruptions, health, and event changes) to subscribers through in-app notifications and browser push notifications, and that lets registered tourists write reviews, publish travel blogs, join discussion forums, save places to a trip plan, and send inquiries and feedback to the tourism office. The third is a role-based administrative console where tourism personnel manage listings, events, alerts, community content, and accounts, and where they view analytics that can be exported as PDF and Excel reports.

The intended users are grouped into five roles. Visitors (guests without an account) can read all public information and alerts. Registered tourists can additionally interact with content and receive personalized alerts. Tourism Officers maintain listings, events, alerts, reviews, inquiries, and tourism analytics. Moderators review community submissions, handle reports, warnings, posting bans, suspensions, and appeals, and maintain forum categories and community guidelines. Administrators manage staff and tourist accounts, roles and permissions, system settings, security policies, audit logs, and database backups.

At a macro level, users access the system through any modern web browser on a phone, or computer. The application is built with Next.js and React and is hosted on a cloud

platform; it stores data in a PostgreSQL database managed through Supabase, which also provides authentication, file storage, and real-time change notifications. When a Tourism Officer publishes an alert, the system writes the alert to every subscriber's notification inbox in a single database operation and requests a push notification to their devices; open pages refresh automatically when tourism content changes. Every privileged action is checked against the user's role on the server and recorded in an audit log.

Background of the Study

Tourism is one of the most information-intensive industries. Travelers decide where to go, where to stay, and what to do largely on the basis of information available online, and destinations increasingly compete on the quality, accessibility, and timeliness of the information they provide. Recent research continues to emphasize this relationship. A systematic literature review found that smart tourism destinations increasingly use technology to support sustainable development, improve visitor experiences, and strengthen the management of tourism information (El Archi et al., 2023). Studies of smart tourism systems likewise show that combining tourism data from several sources in one system gives visitors accurate and timely information while giving

tourism managers data to support decision-making (Qin & Pan, 2023), and that tourists value real-time information such as weather and road conditions (Lei & Kamsin, 2025).

In the Philippines, tourism is recognized as an important component of national development. Republic Act No. 9593, or the Tourism Act of 2009, establishes tourism as an indispensable element of the national economy and provides for the development and promotion of tourism destinations by local government units. More recently, the Department of Tourism's National Tourism Development Plan 2023–2028 emphasizes the improvement of tourism infrastructure, digitalization, visitor services, and destination competitiveness (Department of Tourism, 2023). The Philippine tourism sector has also demonstrated substantial economic recovery and contribution in recent years. According to the Philippine Statistics Authority (PSA), tourism contributed 8.9 percent of the Philippine economy in 2024, with Tourism Direct Gross Value Added reaching ₱2.35 trillion, up from ₱2.12 trillion in 2023 (The Philippine Star, 2025). In its 2025 Philippine Tourism Satellite Account, which revised the earlier series, the PSA reported that tourism contributed 8.1 percent of the economy in 2025, equivalent to ₱2.27 trillion, down from a revised 8.7 percent in 2024 (Manila Bulletin,

2026). [Verify the inbound tourism expenditure of ₱698.46 billion in the PSA release before adding it back.]

The importance of timely tourism information is particularly relevant in the Philippines because destinations are also exposed to typhoons, flooding, and other hazards that can disrupt travel activities. Republic Act No. 10121, or the Philippine Disaster Risk Reduction and Management Act of 2010, provides the national framework for disaster risk reduction and management, while Republic Act No. 10639, or the Free Mobile Disaster Alerts Act, supports the dissemination of disaster-related warnings to the public. For tourism destinations, the availability of timely and authoritative information can therefore contribute not only to visitor convenience but also to public safety during changing conditions.

Daet, the capital municipality of Camarines Norte, serves as an important tourism and commercial center in the Bicol Region. Its tourism identity is associated with attractions such as Bagasbas Beach, which is known for surfing and other beach activities, as well as cultural events such as the Pinyasan Festival, which celebrates the municipality's pineapple industry. At the provincial level, Camarines Norte has demonstrated significant tourism activity in recent years. In 2024, the province recorded 888,810 domestic

tourist arrivals, placing it among the leading provinces in the Bicol Region in terms of domestic arrivals, although the number of foreign arrivals remained considerably smaller (Bicol Mail, 2025). The province's tourism activity continued into 2025, with the Provincial Tourism Office reporting 96,881 local and 120 foreign visitors in April 2025, followed by 95,907 local and 131 foreign visitors in May 2025 [cite the Provincial Tourism Office source of the April and May 2025 figures]. These figures demonstrate the continued importance of tourism to Camarines Norte and highlight the need for accessible and reliable information for both domestic and international visitors.

However, the municipality's capacity to manage and disseminate tourism information has not necessarily kept pace with the increasing need for accessible, timely, and organized information. Tourists may need to consult social media pages, websites, online maps, unofficial posts, and other scattered sources when planning their visits. The Daet Municipal Tourism Office's reliance on social media as a major channel for public communication can create limitations when tourism information becomes more extensive and requires structured management.

Foremost among these concerns is information fragmentation and inconsistency. When tourism information is

distributed across different platforms, visitors may encounter varying details regarding attraction schedules, operating hours, events, facilities, and available services. This can create confusion and reduce confidence in the information being used to plan a trip. Another concern is outdated content. Social media posts can quickly become buried beneath newer content, while static information may remain visible even after details have changed. Consequently, tourists may have difficulty determining which information represents the municipality's latest official update.

The management of visitor inquiries is another challenge. As the number of visitors and online interactions increases, tourism personnel may receive repeated questions concerning attractions, schedules, events, accommodations, transportation, and local activities. Without a centralized system for organizing these inquiries and providing structured information, staff may need to repeatedly respond to similar questions through comments or private messages. This can consume staff time and make it difficult to ensure that every inquiry receives a timely response.

The absence of a centralized real-time alert mechanism is also a significant concern. Tourism conditions can change quickly because of severe weather, road closures, attraction closures, event cancellations, or other unexpected

circumstances. When official updates are distributed primarily through general-purpose social media channels, there may be no dedicated mechanism for presenting urgent tourism-related information in a prominent, organized, and easily accessible manner. This can affect visitor convenience, trip planning, and potentially visitor safety. For tourism personnel, information management can likewise become a manual and reactive process. Without a centralized dashboard, staff may have to manage attraction information, events, announcements, visitor inquiries, and other tourism content through separate processes. The absence of structured feedback and rating tools can also make it difficult to systematically identify visitor concerns, evaluate satisfaction, determine which attractions receive greater attention, and identify areas requiring improvement. Community interaction may similarly occur across platforms that are not specifically designed around the municipality's tourism information-management requirements.

Existing general-purpose platforms only partially address these needs. Global mapping and review platforms can provide attraction listings and visitor ratings but are not controlled by the local government and are not intended to serve as the municipality's official channel for local advisories. Social media platforms provide broad public reach

but are not specifically designed for structured tourism listings, centralized content management, real-time alert organization, visitor feedback analysis, and tourism-related administrative functions. National tourism platforms provide valuable information about destinations across the Philippines but may not provide the depth of municipal-level information required by visitors planning activities within Daet.

These limitations support the need for a dedicated Tourism Information and Real-Time Alert System for the Municipality of Daet, Camarines Norte. Such a system can centralize official tourism information, organize attractions and events, provide timely announcements and alerts, facilitate visitor inquiries and community engagement, and provide tourism personnel with tools for managing content and analyzing visitor feedback. By bringing these functions together in a single platform, the proposed system can help reduce information fragmentation, improve the accessibility and timeliness of tourism information, and support the Daet Municipal Tourism Office in delivering a more organized and responsive digital tourism service.

Company Profile

The client organization of this study is the **Municipal Tourism Culture and Arts Office (MTCAO)** of the Local Government Unit (LGU) of Daet, Camarines Norte. The office operates under the Municipal Government of Daet and is headed by the Municipal Tourism Officer, who works under the supervision of the Municipal Mayor.

The Municipal Tourism Culture and Arts Office is responsible for supporting the promotion and development of tourism, culture, and arts within the Municipality of Daet. Its functions include promoting local tourism attractions and events, supporting tourism-related programs and activities, providing frontline assistance to visitors, and coordinating tourism-related activities with other municipal offices and stakeholders.

Based on the organizational information provided by the office, the Municipal Tourism Culture and Arts Office was **created in 2014 and fully institutionalized in 2020.** The office is composed of one permanent Tourism Officer and Job Order personnel who provide clerical, promotional, liaison, and frontline services.

Office Address: Municipal Hall, Maharlika Highway, Pamorangon, Daet, Camarines Norte, Philippines

Head of Office

Municipal Mayor: Hon. Rossano "Ronie" C. Valencia

Tourism Officer: Jocelyn Cruz Sarical

Administrative Staff

The office is supported by administrative personnel assigned to specific functional areas:

- **Clerical:** Judiel Naris and Edward Dela Torre
  Responsible for clerical, documentation, and administrative support activities.
- **Layout Artist:** Edward Dela Torre
  Responsible for preparing visual and promotional materials for tourism-related programs, activities, and events.
- **Liaison:** Imee Dañas
  Responsible for assisting with coordination and communication between the tourism office and other offices, organizations, and stakeholders.
- **Frontline Services:** Fourth Carraneja
  Responsible for providing direct assistance and information to visitors and individuals seeking tourism-related services.

Organizational Structure

The organizational structure of the Municipal Tourism Culture and Arts Office follows a hierarchical arrangement.

At the top is the **Municipal Mayor, Hon. Rossano "Ronie" C. Valencia,** who provides executive leadership within the municipal government. The **Tourism Officer, Jocelyn Cruz Sarical,** serves as the head of the tourism office and oversees its tourism-related programs, activities, and services.

Under the Tourism Officer are the administrative personnel who perform specific operational functions. **Judiel Naris and Edward Dela Torre** handle clerical responsibilities, while **Edward Dela Torre** also performs layout artist functions. **Imee Dañas** serves as the liaison personnel, and **Fourth Carraneja** handles frontline services.

This organizational structure reflects the office's operational setup, in which a permanent Tourism Officer is supported by personnel assigned to administrative, promotional, coordination, and visitor-service functions. The arrangement enables the office to manage tourism-related activities and provide services to visitors while coordinating with other municipal offices and stakeholders.

Significance of the Study

The development of Connect Daet is expected to benefit the following stakeholders, listed according to how directly they are affected by the system:

Daet Municipal Tourism Culture and Arts Office (management and Tourism Officers). The system gives the office a single dashboard for managing listings, events, alerts, reviews, and inquiries, and provides analytics on traffic, popular places, ratings, and alert reach that support data-driven planning. It reduces repetitive manual work and enables proactive rather than reactive tourism management.

Tourists and Visitors. The system provides a reliable, centralized, and mobile-friendly source of accurate information about attractions, accommodations, restaurants, businesses, and events. Visitors receive official real-time alerts on safety and event changes, can plan trips, and can read reviews and community advice.

Moderators and System Administrators. Moderators gain structured tools for pre-moderating community content, handling reports, and enforcing community guidelines. Administrators gain controls for accounts, roles, security policies, audit logs, and backups that protect the integrity of the platform.

Local Businesses and Establishments. Hotels, restaurants, and other tourism-related businesses gain visibility through the official places directory, along with verified reviews and the tourism office's public replies.

Local Government Unit of Daet. The system supports the municipality's tourism and e-governance goals, strengthens its destination image, and provides an official channel for public advisories.

Community and Residents. By supporting tourism growth, the system can contribute to local livelihood, and by showcasing local heritage and events, it can foster community pride.

Future Researchers and Developers. The documented design, database structure, security measures, and evaluation results can serve as a reference for similar municipal tourism and alert systems.

Statement of the Problem

Generally, this study addresses the problem that the Daet Municipal Tourism Office lacks a centralized, official, and real-time platform for disseminating tourism information, issuing advisories, engaging with visitors, and managing tourism data, which results in fragmented and outdated information, slow responses to inquiries, and decisions made without reliable data.

Specifically, it seeks to answer the following questions:

1. What are the operational challenges and communication gaps in the current tourism information dissemination and visitor engagement methods of the Municipality of Daet?
2. What technical, functional, and security requirements are needed in developing a real-time tourist information and management system for the Municipality of Daet?
3. How can the Daet Tourist Information, Engagement, and Management System with Real-Time Updates be designed and developed to provide: (a) a centralized and mobile-responsive tourist information portal; (b) real-time alerts and announcements; (c) interactive features such as reviews, blogs, forums, and feedback; and (d) an administrative console with role-based access and analytics?
4. How acceptable is the developed system based on the ISO/IEC 25010 Software Quality Model, particularly in terms of Usability?

Objectives of the Study

The general objective of this study is to develop a functional, secure, and real-time system that enhances tourism information dissemination and stakeholder engagement for the Municipality of Daet.

Specifically, it aims to:

1. Identify the operational challenges and communication gaps in the current tourism information dissemination and visitor engagement methods of the Municipality of Daet.
2. Analyze the technical, functional, and security requirements of a real-time tourist portal, specifically the alert and push notification architecture, mobile-responsive standards, community moderation rules, and role-based access controls.
3. Design and develop the Daet Tourist Information, Engagement, and Management System with Real-Time Updates, integrating a mobile-responsive places and events portal, a real-time alert broadcast module, interactive community features (reviews, blogs, forums, feedback, and inquiries), and a role-based administrative console with analytics.
4. Evaluate the Usability of the system based on the ISO/IEC 25010:2023 software quality model using a five-point Likert scale questionnaire, in terms of appropriateness recognizability, learnability, operability, user error protection, user engagement, inclusivity, user assistance, and self-descriptiveness, with emphasis on mobile use by tourists.

Scope and Limitations

This study covers the design, development, and evaluation of Connect Daet, a web-based Progressive Web Application accessible through modern web browsers on desktop computers, tablets, and mobile phones. The system provides tourism information for the Municipality of Daet, Camarines Norte; issues real-time alerts; supports moderated community engagement; and supports tourism information management through a role-based administrative console. It uses PostgreSQL through Supabase as its database, Supabase Authentication for sign-in, Supabase Storage for photos and videos, and Supabase Realtime (WebSocket subscriptions) for live updates.

The system serves five user roles. Visitors can browse and search attractions, accommodations, restaurants, local businesses, and events; filter them by category, barangay, and rating; view photos, maps, contact details, and directions; read blogs, forum threads, and reviews; view current and past public alerts; read the About, Contact, and Community Guidelines pages; and subscribe a device to public alerts. Registered tourists can additionally register, sign in, recover passwords, edit profiles, save favorites and plan trip dates, write and edit reviews, publish blogs, take part in forums, react to and report content, choose alert

categories, use a notification inbox, send inquiries and feedback and track their status, and earn points and badges.

Tourism Officers manage listings (with photos, map coordinates, operating hours, entrance fees, contact details, categories, tags, featured status, publishing, and archiving), events and festivals, real-time alerts (immediate or scheduled, targeted to all subscribers or to one category, edited, cancelled, or closed with an "all clear" message, with delivery and reach counts and reusable templates), public replies to reviews, inquiry responses, and tourism analytics with PDF and Excel export. Moderators manage pre-moderation queues for blogs, forum posts, and comments; reports; warnings, posting bans, temporary suspensions, and appeals; forum categories and threads; official announcements; and the community guidelines. Administrators manage staff and tourist accounts, roles and per-person permissions, system settings (site identity, logo, contact and municipality information, alert and place categories, password policy, session timeout, login lockout, notification and email settings), audit logs, failed sign-in records, and database backup and restore.

The geographical coverage is limited to tourism destinations, establishments, and events within the Municipality of Daet.

The study is subject to the following limitations. Platform and distribution. The system is distributed only as a Progressive Web Application and is not published as a native application in the Google Play Store or Apple App Store. It is not designed for kiosks, smart TVs, or wearable devices. Compatibility is limited to current versions of modern browsers, and older browsers may not support all features. Commerce and transactions. The system does not include online booking or room reservations, electronic payment, ticket sales, transport reservations, tour package sales, vouchers, or price comparison. Accommodation and establishment listings are informational only; availability and rates must be confirmed directly with each establishment. Local businesses cannot manage their own listings or respond to reviews; listings and official replies are maintained only by the tourism office.

Artificial intelligence and intelligent features. The system does not include an AI assistant or chatbot, AI-generated itineraries or trip recommendations, image recognition, voice search or voice commands, or automatic translation of user-generated content. Content moderation relies on human moderators supported by a configurable list of prohibited words; it does not use AI to detect harmful text or images. Analytics are descriptive (counts, trends,

and averages) and do not include predictive analytics or forecasting of tourist arrivals.

Maps and directions are provided through Google Maps and require an internet connection; the system does not provide offline maps, turn-by-turn navigation within the application, real-time location tracking of users, geofenced alerts, or location-based check-ins. It does not include virtual tours, 360-degree views, augmented reality, audio guides, or QR-code markers at attractions. Location is requested only optionally during account onboarding, and its accuracy depends on the user's device.

Alerts are delivered only through the in-app notification inbox and browser push notifications. The system does not send SMS text messages, automated voice calls, or alert emails, and it is not connected to national alert systems such as the Emergency Cell Broadcast System, PAGASA, or the NDRRMC; advisories are encoded manually by authorized tourism personnel. Browser push notifications depend on the user granting permission and on browser support; on iPhones and iPads, the site must first be added to the Home Screen. Alerts are always delivered to the in-app inbox, but push delivery to a device is not guaranteed. Inquiries are answered asynchronously by the tourism office; there is no live chat, video call, or telephone integration with the office. Emails

depend on the SendGrid service, maps and directions depend on Google Maps, and weather information appears only on the administrator dashboard and depends on the OpenWeather service; public weather forecasts, tide charts, and live surf or crowd conditions are not provided.

The system does not integrate with the Department of Tourism's systems or accreditation records, the eTravel platform, other local government information systems, hotel property-management systems, or third-party travel platforms. It does not offer sign-in through social media or Google accounts, and it does not share content to social media automatically.

Accounts are protected by passwords, configurable password rules, login lockout, and session timeouts, but the system does not offer two-factor authentication, biometric sign-in, or single sign-on. The provision of server hardware, paid hosting plans, and domain registration is outside the scope; the system relies on cloud services (Supabase for the database, authentication, and storage, and a cloud host for the application), so storage, bandwidth, uptime, and connection limits depend on the plans used. Database backups cover records only; uploaded photos and videos must be backed up separately.

Full functionality, including real-time updates and alerts, requires an internet connection; offline access is limited to static assets cached by the PWA, and actions taken while offline are not saved for later submission.

The accuracy and timeliness of information depend on the tourism office personnel who maintain it, and user-generated content is published only after moderator approval. The interface is available in English, Filipino, and Bikol for navigation and settings, but user-generated content and most page content are not translated, and other languages are not supported. The system was designed to follow the Web Content Accessibility Guidelines 2.1, but it has not been formally certified for accessibility, and features such as sign-language content or screen-reader-optimized audio descriptions are not provided.

The system covers only tourism destinations, establishments, and events within the Municipality of Daet and does not include neighboring municipalities or provinces. Evaluation is limited to the Usability characteristic of ISO/IEC 25010, measured through a five-point Likert scale questionnaire; other quality characteristics such as performance efficiency, reliability, and security were verified through developer testing but were not rated by respondents. The evaluation does not measure the system's

long-term effect on tourist arrivals, visitor satisfaction, or the local economy.

Finally, the system is an information and advisory platform and does not replace official emergency response channels; users are advised to contact local authorities or the national emergency hotline 911 during emergencies.

Definition of Terms

The following terms are defined operationally as they are used in this study:

Administrative Console. The restricted part of the system at /info/admin used by Administrators, Tourism Officers, and Moderators. Each role sees only the menus and actions its permissions allow.

Alert. An official advisory issued by a Tourism Officer with a title, message, category, severity (info, advisory, warning, or critical), and affected area, delivered to subscribers' inboxes and devices, either immediately or at a scheduled time.

All Clear. A follow-up message attached to a sent alert that tells subscribers the situation has ended.

Audit Log. A database record of every privileged action, such as role changes, deletions, sent alerts, and setting changes, including who performed it and when.

Engagement. The interactive features that let users take part in the platform: reviews, blogs, forums, reactions, reports, inquiries, and feedback.

Gamification. The use of points, badges, and rewards that registered tourists earn for approved participation.

Listing. A record in the places directory that describes an attraction, accommodation, restaurant, or local business.

Pre-moderation. The rule that blogs, forum posts, replies, and comments submitted by tourists are held as pending and become public only after a Moderator approves them.

Progressive Web Application (PWA). A website that can be installed on a device's home screen, caches static assets, and can receive push notifications, while still running in a web browser.

Push Notification. A message sent through the browser's push service to a subscribed device even when the site is not open.

Reach. The number of alert recipients who opened the alert in their inbox.

Real-Time Update. An automatic refresh of an open page shortly after the underlying data changes, using Supabase Realtime notifications, with a 30-second reconciliation fallback.

Role-Based Access Control (RBAC). The security rule set that grants each user role a fixed set of permissions, enforced on pages, on server APIs, and in the database.

Row-Level Security (RLS). Database policies in PostgreSQL that decide which rows each signed-in or anonymous user may read or change.

Supabase. The cloud platform that provides the system's PostgreSQL database, authentication, file storage, real-time notifications, and serverless functions.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of relevant literature and previous studies that form the theoretical and empirical foundation for the development of the Daet Tourist Information, Engagement, and Management System with Real-Time Updates. The discussion is organized into three sections: foreign and local literature, foreign and local studies, and a synthesis that connects these findings to the present project.

Related Literature

Foreign Literature

The digital transformation of tourism management has increasingly emphasized the use of information and communication technologies to improve destination management, visitor experiences, and decision-making. El Archi et al. (2023) conducted a systematic literature review of smart tourism destinations and found that recent research increasingly examines the role of technology in achieving sustainable tourism development. Their review highlights the growing importance of digital technologies in managing tourism resources, improving visitor services, and supporting destination-level decision-making. This supports the development of a destination-managed digital platform such as

Connect Daet, which aims to organize tourism information and provide timely updates to visitors.

Qin and Pan (2023) designed a smart tourism management system based on multisource data visualization and knowledge discovery. Their system consists of three major modules: data collection, data visualization, and knowledge discovery. The study demonstrates how information gathered from multiple sources can be processed and visualized to generate statistical information that supports tourism management and decision-making. This is relevant to Connect Daet because the proposed system consolidates tourism information, events, reviews, inquiries, and other records into a centralized platform that can be used by tourism personnel for management and monitoring.

Recent tourism research has also emphasized the importance of timely and context-aware information. Lei and Kamsin (2025) examined the application of knowledge graph techniques to tourism recommendation systems and discussed the limitations of conventional systems in personalization, contextual awareness, and real-time processing. Their work emphasizes the importance of tourism technologies that can respond to changing circumstances and contextual information. This relates to Connect Daet's real-time alert functionality, which is designed to provide official information about

changes that may affect tourists, including weather conditions, closures, schedule changes, and other tourism-related advisories.

Kovačić and Car (2023) examined destination management from the perspective of tourism authorities and investigated the role of technology and data-driven approaches in destination management. Their study is relevant to the present research because Connect Daet is designed specifically for use by a municipal tourism office. While international smart tourism research often focuses on large destinations and advanced technologies, the present study applies similar principles to a smaller local-government setting where centralized information management and accessible digital services are needed.

Local Literature

The Philippine government has increasingly promoted the digitalization of tourism and public services. The Department of Tourism's National Tourism Development Plan 2023–2028 emphasizes improving tourism competitiveness, visitor services, digitalization, and the use of technology in tourism development. These directions demonstrate the national government's recognition of digital technology as an important component of modern tourism management.

The Department of Tourism also continued to strengthen digital tourism promotion through partnerships with private-sector digital platforms. In 2025, the Department partnered with Klook to promote Philippine destinations and tourism experiences through a digital platform (ABS-CBN News, 2025). Such initiatives demonstrate the growing role of digital channels in making tourism destinations more accessible to potential visitors. However, these platforms primarily support destination promotion and booking rather than the day-to-day information-management and alert requirements of a municipal tourism office.

Digitalization has likewise expanded within Philippine government services. The eGovPH Super App, launched by the Department of Information and Communications Technology in 2023, seeks to consolidate government services into a single digital platform (ABS-CBN News, 2023). The initiative demonstrates the potential of centralized digital government services to simplify public access to information and transactions. However, a national government application serves a broader range of government functions and does not specifically address the tourism information, community engagement, and real-time advisory requirements of a municipal tourism office.

The Asian Development Bank (2024) has also highlighted differences in digital and smart-tourism readiness among Southeast Asian countries. The findings indicate that countries in the region differ in their capacity to adopt and implement technologies supporting smart tourism. This

suggests that strengthening digital capabilities at the local-government level remains important, particularly for municipalities seeking to improve the accessibility and management of tourism information.

Cardaña (2025) examined stakeholder perspectives regarding a proposed tourism databank for Panglao Island, Bohol. The study emphasized the importance of system reliability, security, usability, accessibility, research integrity, and economic impact in the development of a digital tourism solution. The study is relevant to Connect Daet because both focus on applying digital information systems to tourism management within a Philippine local-government context. It also highlights the importance of considering stakeholder requirements and system quality when developing tourism technologies.

Related Studies and Systems

Foreign Studies and Systems

Cui (2025) proposed the Urban Tourism Evaluation System (UTES), which integrates techniques such as collaborative filtering, long short-term memory neural networks, fuzzy logic, big-data analysis, and K-means clustering for urban tourism management. The study demonstrates the potential of artificial intelligence and data analytics in tourism

forecasting and recommendation. However, its technological requirements and dependence on large datasets differ from Connect Daet, which focuses on providing reliable official information, real-time alerts, community engagement, and administrative tools appropriate for a municipal tourism office.

Hernández-Cabrera, Plácido-Castro, and Bulchand-Gidumal (2024) presented the TOURETHOS platform, which supports destination management through the collection and analysis of tourist-movement data. The system demonstrates how digital technologies can provide tourism managers with information about visitor movement and destination behavior. Its emphasis on data-driven destination management is similar to Connect Daet's use of analytics; however, Connect Daet focuses on tourism information dissemination, official alerts, inquiries, reviews, and community engagement rather than primarily tracking physical tourist movement.

Dube and Humbani (2024) examined the factors influencing engagement with mobile travel applications using an extended DeLone and McLean Information Systems Success Model. Their findings emphasize the importance of information quality and system quality in influencing user engagement. These findings support Connect Daet's emphasis on accurate tourism

information, an accessible interface, reliable system performance, and organized services for tourists.

Omar, Tiwari, and Saad (2025) investigated technology acceptance in smart destinations using an extended UTAUT framework that considered privacy and safety risks. Their study emphasizes the importance of performance expectancy, effort expectancy, and facilitating conditions in determining users' intention to use smart tourism technologies. The study also highlights the relevance of privacy and safety concerns. These findings support the inclusion of privacy-conscious features in Connect Daet, particularly in relation to user accounts, analytics, inquiries, and community participation.

Local Studies and Systems

Arruejo and Arruejo (2025) developed and evaluated Viajefy, a tourism information system for Ilocos Sur that uses the Term Frequency–Inverse Document Frequency (TF-IDF) algorithm to support tourism recommendations. The system demonstrates the feasibility of developing a government-oriented tourism information platform in the Philippines. Similar to Connect Daet, it provides tourism-related information through a digital platform. However, Connect Daet extends the concept by incorporating real-time official alerts, moderated community features, tracked inquiries,

official responses to reviews, and role-based administrative functions.

Rivas (2026) developed a barangay information management system that centralized resident information, records, and report generation while incorporating automated mobile messaging for real-time communication. The study demonstrates the benefits of centralized information management and automated communication within a local-government environment. Although the system focuses on barangay administration rather than tourism, its use of centralized records and real-time communication provides relevant support for the design of Connect Daet.

Padilla ([year]) developed Barangay Connect, a management information system designed to digitize barangay records, automate information processing and report generation, and provide SMS notifications for real-time communication. The study demonstrates how a local-government information system can combine administrative functions with real-time public communication.

Aliling et al. (2025) examined the digital transformation of Barangay Trinidad through an Electronic Barangay Information Management System. The study focused on automating resident profiling, certificate issuance, and report generation and emphasized the importance of continuous system improvement and personnel training. Although its

application is in barangay governance rather than tourism, it demonstrates the potential of digital systems to improve the efficiency and organization of local-government services.

Tadeo (2024) examined tourism in Cavite Province and its relationship with local-government income using statistical analysis. The study demonstrates the economic importance of tourism to local government units and supports the development of systems that can help local governments manage and promote tourism-related information and activities. Connect Daet builds on this perspective by providing a digital platform that supports tourism promotion, information management, visitor engagement, and administrative monitoring.

Synthesis

The reviewed literature and studies demonstrate the increasing importance of digital technologies in tourism management. Smart tourism research emphasizes the use of technology and data to improve destination management, visitor services, and decision-making. Studies such as El Archi et al. (2023) and Qin and Pan (2023) demonstrate the value of smart tourism technologies and multisource information management, while recent studies emphasize the

importance of timely information, system quality, privacy, and user acceptance.

Foreign systems demonstrate the potential of advanced technologies for tourism forecasting, visitor-flow analysis, recommendation, and smart-destination management. However, many of these systems require sophisticated technologies, extensive datasets, and infrastructure that may not be readily available to a small municipal tourism office. Local studies, meanwhile, demonstrate that centralized information management and real-time communication can be effectively applied within Philippine local-government settings.

The reviewed Philippine tourism systems also reveal a gap. Existing tourism platforms generally emphasize destination promotion, recommendations, or tourism information, while local-government information systems tend to focus on administrative records and communication. Few systems b within a single platform.

Connect Daet addresses this gap by integrating these functions into one locally managed system. The proposed system provides a structured directory of tourism attractions, accommodations, restaurants, and other establishments; event information; map-based location information; category-targeted real-time alerts; moderated community features; visitor reviews and official replies;

tracked inquiries; and role-based administrative functions with analytics and audit records.

Therefore, the proposed system does not simply replicate an existing tourism website or social media page. Instead, it combines tourism information dissemination, visitor engagement, real-time communication, and tourism-office management into a single platform designed specifically for the operational requirements of the Municipality of Daet.

Feature	Viajefy	Barangay Information Systems	National Government Platforms	Connect Daet
Official government-managed tourism information	Yes	No	Partial	Yes
Tourism attraction and establishment directory	Yes	No	Partial	Yes
Search and category filtering	Yes	Partial	Partial	Yes
Real-time public notifications	No/limited	Yes	Partial	Yes
Tourism-specific real-time alerts	No	No	Partial	Yes
Moderated community features	No	No	No	Yes
Visitor reviews and official replies	Limited	No	Partial	Yes
Tracked visitor inquiries	No	Partial	Partial	Yes
Role-based administration	Partial	Yes	Yes	Yes

Tourism analytics	Partial	Partial	Partial	Yes
Exportable administrative reports	Partial	Yes	Partial	Yes

Table 1. Comparison of Related Systems and Connect Daet

The comparison shows that existing systems provide useful individual functions but do not fully address the combined requirements identified for the Daet Municipal Tourism Culture and Arts Office. Connect Daet therefore integrates the complementary functions identified across the reviewed systems into a single municipal tourism platform.

Theoretical Framework

This study is anchored on established information-system and technology-acceptance theories. Although some of the foundational models were originally developed before 2022, they remain relevant because they provide the theoretical basis for evaluating users' acceptance, system quality, and usability.

Technology Acceptance Model (TAM)

The Technology Acceptance Model developed by Davis (1989) proposes that perceived usefulness and perceived ease of use influence a user's intention to use a technology. In Connect Daet, perceived usefulness is associated with features such as accurate tourism listings, real-time alerts,

event information, and tracked inquiries. Perceived ease of use is associated with the system's mobile-responsive interface, organized navigation, search functionality, and simple interaction processes.

Recent tourism technology research continues to support the relevance of technology-acceptance concepts. Omar et al. (2025), for example, found that performance expectancy and effort expectancy influence users' behavioral intention toward smart technologies in destination settings. These concepts correspond closely to the usefulness and ease-of-use principles underlying TAM.

DeLone and McLean Information Systems Success Model

The DeLone and McLean Information Systems Success Model (DeLone & McLean, 2003) provides a framework for evaluating information-system success through dimensions including information quality, system quality, service quality, use, user satisfaction, and net benefits.

For Connect Daet, **information quality** is addressed through tourism-office-managed content and moderation; **system quality** is addressed through reliability, security, responsiveness, and real-time functionality; and **service quality** is supported through visitor inquiries, official replies, and tourism-office assistance. These factors are

expected to contribute to user satisfaction and the perceived benefits of the system.

Dube and Humbani (2024) provide recent tourism-related support for the relevance of information and system quality in influencing user engagement with travel applications.

ISO/IEC 25010:2023 Product Quality Model

The study uses the **ISO/IEC 25010:2023 Product Quality Model** as the basis for evaluating the usability of Connect Daet. The 2023 revision identifies **Interaction Capability** as a product-quality characteristic and provides sub-characteristics that include appropriateness recognizability, learnability, operability, user error protection, user engagement, inclusivity, user assistance, and self-descriptiveness.

The model provides a structured basis for determining whether Connect Daet can be effectively understood, learned, operated, and used by its intended users. It complements TAM because perceived ease of use can be examined through measurable characteristics of the system's interaction capability.

Theoretical Paradigm

The theoretical framework of the study integrates three perspectives:

1. **Technology Acceptance Model (TAM)** — explains users' perceived usefulness and ease of use.
2. **DeLone and McLean Information Systems Success Model** — provides dimensions for examining information quality, system quality, service quality, user satisfaction, and system benefits.
3. **ISO/IEC 25010:2023** — provides the quality model used to evaluate the system's interaction capability/usability.

These frameworks collectively guide the design and evaluation of Connect Daet. TAM focuses on whether users perceive the system as useful and easy to use, the DeLone and McLean model addresses the quality and effectiveness of the information system, and ISO/IEC 25010:2023 provides measurable product-quality characteristics for the system evaluation.

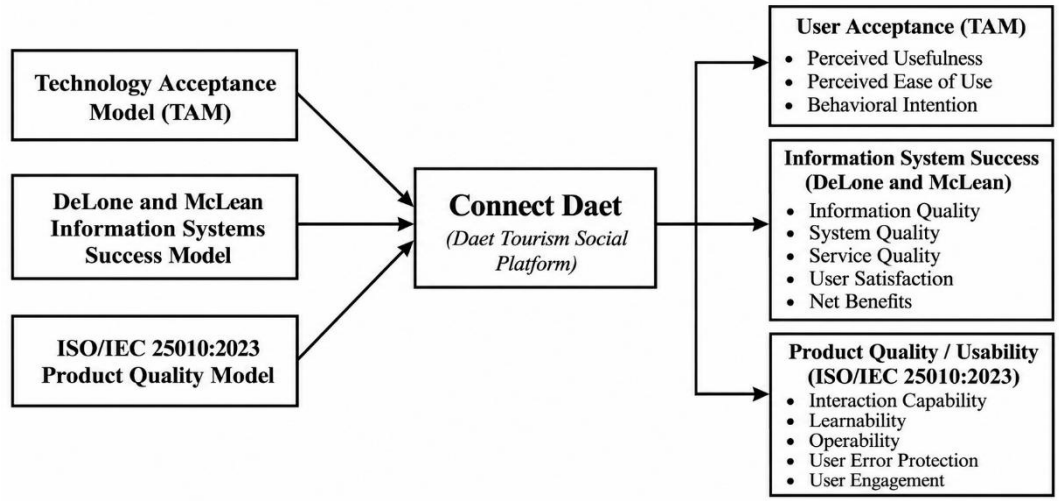


Figure 1. Theoretical Paradigm

Conceptual Framework

The study follows the **Input–Process–Output (IPO) model.** The input includes the knowledge, software, and hardware requirements for developing Connect Daet. Knowledge requirements cover the tourism information-management practices of the Daet Municipal Tourism Culture and Arts Office, existing information-dissemination and visitor-engagement needs, ISO/IEC 25010:2023, and data-privacy requirements under Republic Act No. 10173. Software requirements include Next.js, React, Tailwind CSS, Supabase, Web Push, SendGrid, Google Maps, OpenWeather, Git, and Visual Studio Code. Hardware requirements include development computers, mobile devices for testing, and reliable internet connectivity.

The system follows an **Agile Scrum** approach involving requirements gathering, sprint planning, system and interface design, development, testing, deployment, and evaluation. Requirements are gathered through interviews and observation, while system design includes use-case diagrams, data-flow diagrams, entity-relationship diagrams, and user-interface designs. Testing covers functionality, usability, security, compatibility, and black-box testing. The completed system is evaluated using an ISO/IEC 25010:2023-based questionnaire involving tourists, tourism-office personnel, and IT experts.

The expected output is the **Daet Tourist Information, Engagement, and Management System with Real-Time Updates),** a secure and mobile-responsive Progressive Web Application. It provides

tourism information, attraction and event details, real-time alerts, moderated community features, reviews and inquiries, and role-based administrative functions with analytics and audit capabilities. The study also produces the **system evaluation results** based on the selected ISO/IEC 25010:2023 quality criteria.

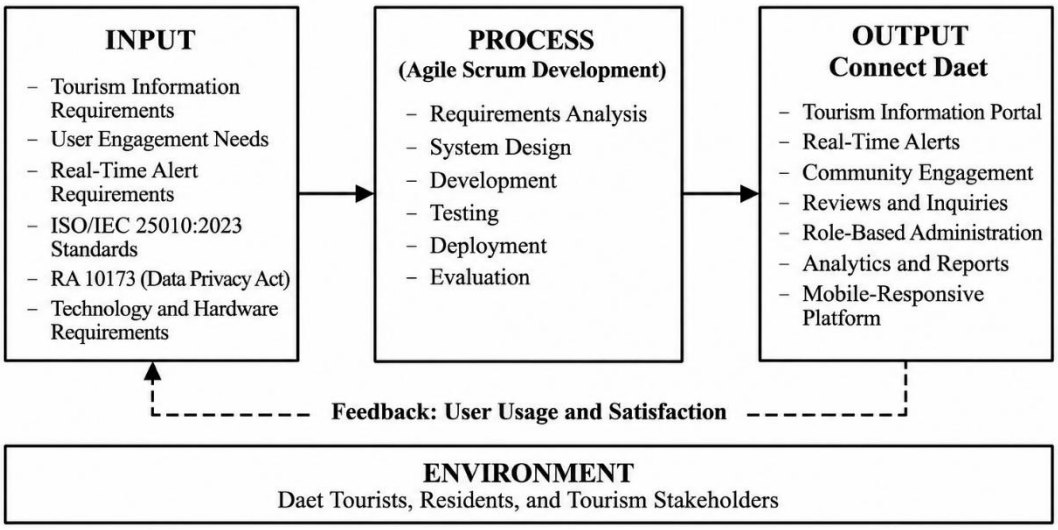


Figure 2. Input–Process–Output Conceptual Framework

CHAPTER 3

METHODOLOGY

This chapter presents the operational framework and engineering approach used to develop, implement, and validate the study. It details the procedures used to translate requirements into a working system and to evaluate its quality.

Research Design

This study uses a descriptive-developmental research design. The developmental part covers the analysis, design, construction, and testing of Connect Daet using an established software development life cycle. The descriptive part has two stages. First, it gathers and describes the current tourism information practices and problems in Daet through interviews, observation, and a needs-assessment survey of residents and tourists. Second, it describes the respondents' evaluation of the developed system using a usability questionnaire based on ISO/IEC 25010:2023. This combination is appropriate because the study must both produce a working system and provide measurable evidence of its quality.

Participants of the Study

The study involved two groups of participants, one for each stage of the descriptive part of the research.

Needs-assessment survey respondents. Twenty (20) residents and tourists of Daet answered the needs-assessment survey, which was distributed online through Google Forms from September to November 2026. Convenience sampling was used: the survey link was shared with people who live in, study in, or have visited Daet, and all valid responses received were included. The results were used to answer SOP 1 and SOP 2.

Respondent type	Frequency	Percentage
Local resident of Daet	12	60.00
Local tourist (from other parts of the Philippines)	7	35.00
Foreign tourist	1	5.00
Total	**20**	**100.00**

Table 2. Distribution of Needs-Assessment Survey Respondents

Usability evaluation respondents. The developed system was evaluated by forty-two (42) respondents divided into three groups, as shown in the table below. The tourism office personnel were selected through total enumeration, the IT experts through purposive sampling, and the tourists and

residents through convenience sampling. The results were used to answer SOP 4.

Group	Description	Sampling technique	Number
Tourism office personnel	Municipal Tourism Officer or staff who will use the administrative console	Total enumeration / purposive	1
IT experts	IT professionals with at least 2 years of experience in systems administration or information system	Purposive	2
Tourists and residents	Visitors and residents of Daet aged 18 and above who use a smartphone or computer	Convenience	39
Total			42

Table 3. Distribution of Usability Evaluation Respondents

Data Gathering Techniques

Interview. Semi-structured interviews with the Municipal Tourism Officer and local residents and tourist were conducted to identify current practices, problems, and requirements (SOP 1 and SOP 2).

Observation and document analysis. The researchers reviewed the office's Facebook page, existing printed and online materials, and inquiry records to document information gaps.

Needs-assessment survey. An online questionnaire was used to gather the respondents' profile, their current sources of tourism information, the problems they experienced with information and feedback, and the features and information they expect from a tourism platform. Most items were multiple choice; some allowed more than one answer or asked for the top three choices, and one item was open-ended. Items about an emoji-based feedback system were also included in the form, but they were excluded from this study because that feature is outside the scope of the developed system.

Usability questionnaire. A researcher-made questionnaire based on the Usability (Interaction Capability) characteristic of ISO/IEC 25010:2023 was used to evaluate the system (SOP 4). Part 1 asks for the respondent's name (optional), role (tourist or visitor, local resident, content creator, administrator, or other), and the date of evaluation. Part 2 contains eleven (11) statements grouped into three areas: (A) Understandability and Learnability, (B) Operability and Real-Time Interaction, and (C) Interface, Accessibility, and Error Recovery. Each statement is rated on

a five-point Likert scale (Likert, 1932) from 5 (Strongly Agree) to 1 (Strongly Disagree). Part 3 contains three open-ended questions: which parts of the system were easiest or most difficult to use, whether any instructions, controls, or real-time updates were unclear, and what changes would make the system easier or more comfortable to use. The table below shows how each statement corresponds to the usability sub-characteristics of ISO/IEC 25010:2023. The questionnaire is attached in Appendix B.

Area	**Statement**	**Sub-characteristic**
A. Understandability and Learnability		
A1	The system's purpose and available features are clear to me.	Appropriateness recognizability
A2	I can easily find tourist information, such as destinations, events, and announcements.	Appropriateness recognizability
A3	The menus, labels, and instructions make it easy to understand how to use the system.	Self-descriptiveness

Area	Statement	Sub-characteristic
A4	I was able to learn how to use the system without needing extensive assistance.	Learnability
B. Operability and Real-Time Interaction		
B1	I can complete the tasks I need—such as searching for information or engaging with content—without unnecessary steps.	Operability
B2	The system's controls and navigation behave consistently across its pages.	Operability
B3	Real-time updates and notifications are easy to notice and understand.	Self-descriptiveness
B4	Forms and interactive features provide clear guidance about what information is needed.	User assistance
C. Interface, Accessibility, and Error Recovery		

Area	Statement	Sub-characteristic
C1	The text, colors, layout, and buttons make the system comfortable to use.	User engagement
C2	The system is easy to use on the device I used, including a mobile phone, tablet, or computer.	Inclusivity
C3	When I make an error, the system explains the problem and helps me correct it.	User error protection

Table 4. Questionnaire Areas and ISO/IEC 25010:2023 Usability Sub-characteristics

System testing. Unit tests, black-box functional tests, role-permission tests, and security checks were performed during development (see Chapter 4, Development and Testing).

Scale	Range	Verbal interpretation
5	4.21 – 5.00	Strongly Agree
4	3.41 – 4.20	Agree
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Disagree
1	1.00 – 1.80	Strongly Disagree

Table 5. Likert Scale and Verbal Interpretation

Sources of Data

Primary sources. Interview responses of the tourism office personnel and of local residents and tourists; needs-assessment survey responses of residents and tourists; usability questionnaire responses of the tourism office personnel, IT experts, and tourists and residents; and test results recorded by the researchers.

Secondary sources. Books, journal articles, standards (ISO/IEC 25010:2023, OWASP Top 10, WCAG 2.1), Philippine laws (RA 9593, RA 7160, RA 10173, RA 10121, RA 10639), Department of Tourism and Philippine Statistics Authority data for 2022–2025, tourist arrival records of the Daet Municipal Tourism Office, and technical documentation of the frameworks and services used (Next.js, Supabase, Web Push).

Statistical Treatment of Data

Frequency and percentage. Used to describe the profile of the respondents and their answers in the needs-assessment survey.

$$P = (f / n) \times 100$$

where P is the percentage, f is the frequency, and n is the total number of respondents.

For items where a respondent could choose more than one answer or the top three choices, each percentage was computed against the total number of respondents (n = 20). For this reason, the percentages of these items may add up to more than 100 percent.

Ranking. Used to arrange the answers to multiple-response items from the most to the least frequently chosen. Answers with the same frequency were given the average of the ranks they occupy.

Weighted mean. Used to determine the respondents' rating of each usability statement, each sub-characteristic, and Usability as a whole.

$$WM = \Sigma(f \times w) \; / \; n$$

where *WM* is the weighted mean, *f* is the frequency of each response, *w* is the weight of the response (1 to 5), and *n* is the number of respondents. The result is interpreted using Table 5. Data were tabulated and computed using Microsoft Excel.

Software Development Methodology

The system was developed using **Agile Scrum** (Schwaber & Sutherland, 2020). Scrum organizes work into short, fixed-length sprints, each producing a working increment that is reviewed. It was chosen because requirements from the tourism

office were refined over time, and each module could be demonstrated and adjusted before the next was built.

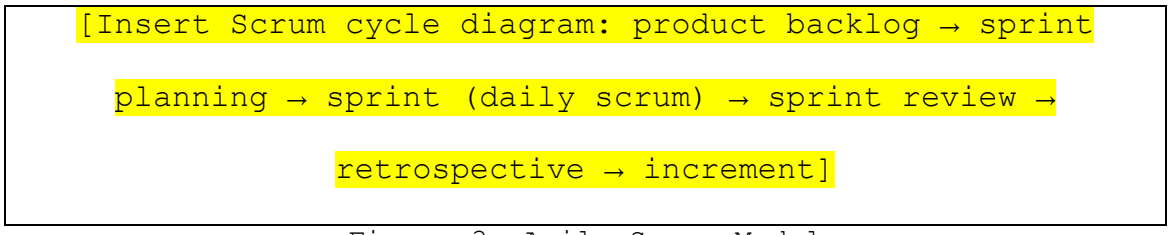


Figure 3. Agile Scrum Model

Requirements Gathering and Planning

Interviews and observations identified the problems described in Chapter 1. Requirements were written as a product backlog grouped by user role (Visitor, Registered Tourist, Tourism Officer, Moderator, Administrator) and prioritized with the tourism office.

Sprints

The backlog was delivered in seven sprints:

1. **Sprint 1 – Accounts and roles:** database schema, registration, sign-in, password recovery, and role-based access control for five roles.
2. **Sprint 2 – Places directory and events:** attractions, accommodations, restaurants, and businesses with photos, maps, hours, fees, categories, tags, featured status, and archiving; events with approval; public pages and search.

3. **Sprint 3 – Real-time alerts:** alert categories, composer, scheduling, targeting, templates, "all clear," delivery and reach tracking, and Web Push.
4. **Sprint 4 – Community features:** reviews and ratings, blogs, forums, reactions, favorites, and inquiries.
5. **Sprint 5 – Moderation:** pre-moderation queues, reports, profanity filter, warnings, posting bans, suspensions, appeals, and community guidelines.
6. **Sprint 6 – Analytics:** traffic, devices, popular places, ratings trends, alert reach, staff activity, and PDF/Excel export.
7. **Sprint 7 – Administration:** system settings, security policy, audit logs, failed sign-ins, backup and restore, and hardening.

Testing and Deployment

Each sprint included automated unit tests, lint and build checks, and manual testing by role. A final integration and security review was performed before deployment to the cloud host and Supabase project, followed by user evaluation.

Project Timeline

Activity	**March-April**	**May-June**	**July**	**August**	**September**	**October**
Requirements gathering	■					
System design	■	■				
Sprints 1–2		■				
Sprints 3–4			■			
Sprints 5–6				■		
Sprint 7 and hardening					■	
Testing and evaluation					■	■
Documentation and deployment						■

Table 6. Gantt Chart

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the findings of the study according to the statement of the problem, the technical background and design of the system, its development and testing, the prototype, and the implementation plan and results.

FINDINGS OF THE STUDY

Answer to SOP 1. The operational challenges and communication gaps in the current tourism information dissemination and visitor engagement methods of the Municipality of Daet

This section presents the results of the needs-assessment survey answered by 20 residents and tourists, together with the interview and observation results. The profile of the respondents is presented first.

Profile	Frequency	Percentage
Age		
18–25	18	90.00
26–35	1	5.00
36–45	1	5.00

Profile	Frequency	Percentage
Gender		
Female	11	55.00
Male	9	45.00
Number of visits to Daet		
More than three times	18	90.00
Two to three times	1	5.00
First time	1	5.00
Tourist spot visited most recently		
Bagasbas Beach	12	60.00
SM City Daet	5	25.00
First Rizal Monument	2	10.00
Daet Municipal Park	1	5.00

Table 7. Profile of the Needs-Assessment Survey Respondents

Most of the respondents were 18 to 25 years old (90.00%), and slightly more were female (55.00%) than male (45.00%). Almost all of them (90.00%) had been to Daet more than three times, which means that they answered from repeated experience rather than a single visit. Bagasbas Beach was the tourist spot most recently visited by more than half of the respondents (60.00%).

Source of information	Frequency	Percentage	Rank
Facebook	16	80.00	1
Asking friends or family	8	40.00	2
Google Search	2	10.00	3.5
Travel blogs or vlogs	2	10.00	3.5
Daet Municipal Tourism Office	1	5.00	5.5
Personal exploration	1	5.00	5.5

Table 8. Sources of Tourism Information About Daet

Dependence on a single social media channel. Facebook was the main source of tourism information for 80.00% of the respondents, followed by asking friends or family (40.00%). Only one respondent (5.00%) obtained information directly from the Daet Municipal Tourism Office. This shows that information about Daet reaches people mostly through social media posts and word of mouth, where official and unofficial information are mixed and older posts are quickly buried by newer ones.

Indicator and response	Frequency	Percentage
How often outdated or incorrect information is encountered		
Always	3	15.00
Often	3	15.00
Sometimes	8	40.00

Indicator and response	Frequency	Percentage
Rarely	6	30.00
How questions about tourism in Daet are answered		
Search on Google	10	50.00
Ask friends or family	7	35.00
Message the Tourism Office on Facebook	2	10.00
Ask locals	1	5.00
Missed an important announcement (event, road closure, or emergency)		
Yes, a few times	12	60.00
Yes, many times	4	20.00
Not sure	4	20.00

Table 9. Experiences with Tourism Information and Announcements

Inconsistent and outdated information. Seventy percent (70.00%) of the respondents encountered outdated or incorrect tourism information at least sometimes, and 30.00% encountered it often or always. No respondent said that this never happens.

Inquiries handled outside official channels. Half of the respondents (50.00%) answered their questions by searching on Google and 35.00% by asking friends or family. Only 10.00% sent a message to the Tourism Office's Facebook page. Because most questions never reach the office, it cannot see what

visitors commonly ask, and the inquiries that do arrive are not tracked in one place.

No dependable channel for urgent advisories. Eighty percent (80.00%) of the respondents had missed an important announcement about an event, a road closure, or an emergency at least a few times, and 20.00% had missed one many times. This shows that posting advisories on a social media page does not ensure that people see them in time.

Indicator and response	Frequency	Percentage
Has given feedback about a tourist spot in Daet		
Yes	14	70.00
No	6	30.00
Method used to give feedback (multiple responses)		
Verbally	9	45.00
Facebook	9	45.00
Google Form	3	15.00
Paper form	2	10.00
Ease of giving feedback at present		
Easy	12	60.00
Not easy	6	30.00
Rather not say	2	10.00

Indicator and response	Frequency	Percentage
Overall experience during the visit		
Very good	14	70.00
Good	4	20.00
Average	2	10.00
Problems noticed at tourist spots (multiple responses)		
Cleanliness or litter	12	60.00
Poor facilities (comfort rooms, waiting areas)	11	55.00
Safety and security	7	35.00
Overcrowding	7	35.00
Lack of information or signage	3	15.00
None	1	5.00

Table 10. Feedback Practices and Problems Observed at Tourist Spots

No structured feedback or data for decisions. Although 70.00% of the respondents had given feedback about a tourist spot, it was given mostly verbally (45.00%) or through Facebook (45.00%), which leaves no organized record that the office can review. Thirty percent (30.00%) said that giving feedback is not easy, and respondents who had not given feedback explained that they did not know how or had never been asked. At the same time, the respondents reported

problems that the office needs to know about, mainly cleanliness (60.00%) and poor facilities such as comfort rooms and waiting areas (55.00%), followed by safety and security and overcrowding (35.00% each). Most respondents still rated their overall experience as very good (70.00%).

[Add the main points from the interview with the Municipal Tourism Officer and from the review of the office's Facebook page and records, for example: how announcements and inquiries are handled at present, how listings and events are updated, and whether visitor records or feedback are kept. Support each point with a short quote or example.]

These findings confirm the need for a single, officially maintained platform with real-time alerts, tracked engagement, and management tools, which shaped the requirements in SOP 2.

Answer to SOP 2. The technical, functional, and security requirements are needed in developing a real-time tourist information and management system for the Municipality of Daet

The requirements of the system were drawn from the needs-assessment survey, the interviews, the review of related systems in Chapter 2, and applicable laws and standards. The

survey results on the respondents' expected use of a tourism platform are presented first.

Indicator and response	Frequency	Percentage
Importance of mobile phone access		
Very important	18	90.00
Important	2	10.00
Would use a central website or app for Daet tourism		
Yes, definitely	11	55.00
Yes, probably	8	40.00
No, probably not	1	5.00
Preferred way to access the platform		
On mobile phone	19	95.00
No preference	1	5.00
Willing to create an account		
Yes, if it is easy and quick	10	50.00
Yes, but concerned about privacy	8	40.00
No, prefer not to register	2	10.00
Would use community features such as forums		
Yes, definitely	9	45.00
Yes, maybe	8	40.00
No, not interested	3	15.00

Indicator and response	Frequency	Percentage
Points or rewards would encourage participation		
Yes, definitely	13	65.00
Yes, maybe	6	30.00
No, not really	1	5.00
Shares travel experiences or recommendations with others		
Yes, always	12	60.00
Yes, sometimes	6	30.00
No, rarely	2	10.00

Table 11. Expected Use of a Central Tourism Platform

Ninety-five percent (95.00%) of the respondents would use a central website or app for Daet tourism, and the same percentage prefer to access it on a mobile phone. All respondents considered mobile access important, and 90.00% considered it very important. These results require a mobile-first, responsive design that works in a phone browser and can be installed as a progressive web app. Ninety percent (90.00%) are willing to create an account, but 40.00% are concerned about privacy, so registration must be quick, optional for browsing, and protected by clear privacy practices. Community features and rewards also have strong support: 85.00% would use forums, 95.00% said points or

rewards would encourage them to participate, and 90.00% already share their travel experiences with others.

Feature (top three)	Frequency	Percentage	Rank
List of tourist attractions	13	65.00	1
Photos and videos	11	55.00	2
Maps and directions	8	40.00	3
Events calendar	7	35.00	4
Emergency and safety alerts	6	30.00	5
Tourist reviews and ratings	5	25.00	6
Travel tips and guides	4	20.00	7
Transportation information	3	15.00	8
Hotel and restaurant directory	2	10.00	9
Community forum	1	5.00	10

Table 12. Most Useful Features of a Tourism Platform

The list of tourist attractions was the most wanted feature (65.00%), followed by photos and videos (55.00%), maps and directions (40.00%), the events calendar (35.00%), and emergency and safety alerts (30.00%). These features form the core of the public portal: a searchable places directory with photo galleries, map locations and directions, an events page, and an alerts module.

Information (top three)	Frequency	Percentage	Rank
Location or address	19	95.00	1
Entrance fees	11	55.00	2
Photos	8	40.00	3.5
Reviews from other tourists	8	40.00	3.5
Operating hours	6	30.00	5
Accessibility information	3	15.00	6.5
Description	3	15.00	6.5
Contact number	2	10.00	8

Table 13. Information Needed in Attraction Listings

Almost all respondents (95.00%) chose the location or address as the most important information in a listing, followed by entrance fees (55.00%), photos and reviews from other tourists (40.00% each), and operating hours (30.00%). Each listing in the system therefore stores its address and map coordinates, fees, photos, operating hours, contact details, accessibility notes, a description, and visitor reviews. One respondent summarized this need by saying that having the location, operating hours, entrance fees, transportation, and current events in one reliable place would make planning much easier.

Type of attraction (top three)	Frequency	Percentage	Rank
Beaches (e.g., Bagasbas Beach)	17	85.00	1
Parks and nature spots	11	55.00	2
Historical sites	10	50.00	3
Restaurants and cafés	9	45.00	4
Churches	5	25.00	5
Festivals (e.g., Pinyasan Festival)	4	20.00	6
Hotels and resorts	3	15.00	7
Shopping areas	1	5.00	8

Table 14. Types of Attractions to Prioritize

Beaches were the top priority (85.00%), followed by parks and nature spots (55.00%), historical sites (50.00%), and restaurants and cafés (45.00%). The places directory therefore groups listings into attractions, accommodations, restaurants, and shops and services, with managed categories so that the office can feature beaches, nature spots, and heritage sites.

Indicator and response	Frequency	Percentage
Preferred way to receive urgent announcements		
In-app notification	13	65.00
Text message (SMS)	6	30.00

Indicator and response	Frequency	Percentage
Facebook message	1	5.00
How often information should be updated		
As soon as changes happen	9	45.00
Daily	7	35.00
Weekly	2	10.00
Monthly	2	10.00
What would make the information trustworthy		
All of the choices below	9	45.00
Official Tourism Office endorsement	6	30.00
Regularly updated information	3	15.00
Positive user reviews	1	5.00
Clear and complete details	1	5.00

Table 15. Preferred Alert Channel, Update Frequency, and Basis of Trust

Most respondents (65.00%) prefer to receive urgent announcements through in-app notifications, while 30.00% prefer text messages. Eighty percent (80.00%) expect the information to be updated as soon as changes happen (45.00%) or daily (35.00%). For trust, 75.00% chose official Tourism Office endorsement either alone or together with all the other factors. These results require real-time page updates, in-app and push notifications for alerts, and content that is

published and maintained by the tourism office. SMS delivery was not included in this version of the system and is listed among the recommendations in Chapter 5.

Survey finding	System feature
95.00% prefer mobile access; 100.00% consider it important	Mobile-first responsive pages; installable progressive web app
80.00% rely on Facebook; only 5.00% use the Tourism Office directly	One official portal managed by the tourism office; links that can be shared on Facebook
70.00% met outdated information; 80.00% want updates as soon as changes happen or daily	Listings and events edited by staff; real-time page refresh through Supabase Realtime
80.00% missed announcements; 65.00% prefer in-app notifications	Alerts module with categories, scheduling, targeting, notification inbox, and Web Push
Attractions, photos, maps, and events are the most useful features	Places directory with galleries, map and directions, filters, and events page
Location, fees, photos, reviews, and hours are needed in listings	Listing fields for address, map coordinates, fees, hours, contact, and photos; reviews and ratings

Survey finding	System feature
70.00% gave feedback only verbally or on Facebook; 30.00% find it hard	Feedback and inquiry forms with status tracking, office replies, and exportable reports
85.00% would use forums; 95.00% are encouraged by rewards	Blogs, forums, reactions, points, and badges with pre-moderation
40.00% are concerned about privacy when registering	Browsing without an account; hidden private account fields; consent-based analytics; RA 10173 compliance

Table 16. Survey Findings and Corresponding System Features

The requirements identified are summarized below.

Category	Requirement
Functional – Visitors	Browse and search places and events; filter by type, category, barangay, and rating; view photos, maps, directions, hours, fees, and contacts; read blogs, forums, reviews, and alerts; read About, Contact, and Guidelines pages; subscribe a device to alerts.
Functional – Registered tourists	Register, sign in, recover password, edit profile and photo; favorites and trip planner; reviews; blogs; forum threads and replies; reactions and reports; alert category preferences; notification inbox; inquiries and feedback with status and replies.

Category	Requirement
Functional – Tourism Officers	Manage listings and events; compose, schedule, target, edit, cancel, and close alerts; templates; delivery and reach counts; reply to reviews; answer inquiries; analytics with PDF/Excel export.
Functional – Moderators	Pre-moderation queues; reports; edit, hide, delete content; profanity filter; warnings, posting bans, suspensions, appeals; forum categories, pin, lock; announcements; community guidelines; moderation log.
Functional – Administrators	Staff and tourist accounts; roles and per-person permissions; settings; security policy; audit logs and failed sign-ins; backup and restore; system overview and analytics.
Technical	Mobile-responsive PWA; real-time page updates and Web Push; cloud database with storage; PDF/Excel generation; scheduled jobs.
Security	Hashed passwords (Supabase Auth); signed HTTP-only sessions with lifetime and idle timeout; server-side RBAC on every admin API; row-level security; login lockout; rate limits; input sanitization; upload content checks; private account columns hidden; audit logging; RA 10173 compliance.

Table 17. System Requirements

Answer to SOP 3. How can the system be designed and developed to provide a centralized mobile-responsive portal, real-time

alerts, interactive features, and an administrative console with role-based access and analytics?

The system was designed and developed as described in the succeeding sections: Technical Background, Design, Software and Systems Architecture, Development and Testing, and Description of the Prototype. Each of the four required capabilities is implemented as follows: (a) the places directory and event pages under /info/visitor; (b) the alerts module with in-app and push delivery; (c) reviews, blogs, forums, and inquiries with pre-moderation; and (d) the role-based console under /info/admin with analytics.

Answer to SOP 4. The acceptability of the developed system in terms of Usability based on the ISO/IEC 25010:2023 software quality model

The developed system was evaluated by 42 respondents using the usability questionnaire described in Chapter 3. The profile of the respondents is presented first, followed by the results for each area of the questionnaire and a summary.

Age group	Frequency	Percentage
18–25	17	40.48
26–35	12	28.57
36–45	8	19.05

Age group	Frequency	Percentage
46–60	5	11.90
Total	**42**	**100.00**

Table 18. Distribution of Usability Evaluation Respondents According to Age

The largest group of respondents was 18 to 25 years old (17 or 40.48%), followed by those aged 26 to 35 (12 or 28.57%), 36 to 45 (8 or 19.05%), and 46 to 60 (5 or 11.90%). The evaluation therefore included both younger and older users. [Note: the questionnaire in Appendix B does not ask for age. Confirm where these age figures were recorded, add an age item to the form, or remove this table.]

Respondent type	Frequency	Percentage
Residents of Daet	33	78.57
Visitors from other places	9	21.43
Total	**42**	**100.00**

Table 19. Distribution of Usability Evaluation Respondents According to Residence

Thirty-three respondents (78.57%) were residents of Daet and nine (21.43%) were visitors from other places, giving the views of both people who know the municipality well and people who rely more on published information.

A. Understandability and Learnability

Statement	5	4	3	2	1	WM	Interpretation
1. The system's purpose and available features are clear to me.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
2. I can easily find tourist information, such as destinations, events, and announcements.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
3. The menus, labels, and instructions make it easy to understand how to use the system.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
4. I was able to learn how to use the system without needing extensive assistance.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Overall weighted mean						**[ ]**	**[ ]**

Table 20. Evaluation of the System in Terms of Understandability and Learnability

[Discussion: state the overall weighted mean of Understandability and Learnability and its interpretation, then the statements with the highest and lowest weighted means, and what these mean for the users. Use the computed results from the actual questionnaire responses.]

B. Operability and Real-Time Interaction

Statement	5	4	3	2	1	WM	Interpretation
1. I can complete the tasks I need—such as searching for information or engaging with content—without unnecessary steps.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
2. The system's controls and navigation behave consistently across its pages.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
3. Real-time updates and notifications	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

Statement	5	4	3	2	1	WM	Interpretation
are easy to notice and understand.							
4. Forms and interactive features provide clear guidance about what information is needed.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Overall weighted mean						**[ ]**	**[ ]**

Table 21. Evaluation of the System in Terms of Operability and Real-Time Interaction

[Discussion: state the overall weighted mean of Operability and Real-Time Interaction and its interpretation, then the statements with the highest and lowest weighted means, and what these mean for the users. Use the computed results from the actual questionnaire responses.]

C. Interface, Accessibility, and Error Recovery

Statement	5	4	3	2	1	WM	Interpretation
1. The text, colors, layout, and buttons	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

Statement	5	4	3	2	1	WM	Interpretation
make the system comfortable to use.							
2. The system is easy to use on the device I used, including a mobile phone, tablet, or computer.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
3. When I make an error, the system explains the problem and helps me correct it.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Overall weighted mean						**[ ]**	**[ ]**

Table 22. Evaluation of the System in Terms of Interface, Accessibility, and Error Recovery

[Discussion: state the overall weighted mean of Interface, Accessibility, and Error Recovery and its interpretation, then the statements with the highest and lowest weighted means, and what these mean for the users. Use the computed results from the actual questionnaire responses.]

Summary of the Usability Evaluation

Evaluation area	Weighted mean	Interpretation
Understandability and Learnability	[ ]	[ ]
Operability and Real-Time Interaction	[ ]	[ ]
Interface, Accessibility, and Error Recovery	[ ]	[ ]
Overall Usability	**[ ]**	**[ ]**

Table 23. Summary of the Usability Evaluation

[Discussion: state the overall weighted mean of Usability and its interpretation, rank the three areas, and relate the lowest-rated area to the answers in Part 3 of the questionnaire (open-ended comments). Use the computed results from the actual questionnaire responses.]

TECHNICAL BACKGROUND OF THE PROJECT

The system was built with widely used, well-documented open technologies so that it can be maintained by the LGU or future developers.

Layer	Technology	Reason for choice
Front end	Next.js 16 (App Router), React 19, Tailwind CSS 4, Lucide icons	Component-based UI, server and client rendering, responsive utility styling.
Server / API	Next.js Route Handlers (Node.js runtime), Next.js Proxy	API endpoints and request-level session, role, idle-timeout, and maintenance checks in the same codebase.
Database	PostgreSQL via Supabase (111 migrations, 87 tables)	Relational integrity, row-level security, functions and triggers, managed backups.
Authentication	Supabase Auth + signed HTTP-only session cookie	Hashed passwords, email verification, recovery; tamper-proof role in session.
Real time	Supabase Realtime (WebSocket), Web Push (VAPID) via Supabase Edge Function, pg_cron	Live page refresh, device notifications, scheduled alert delivery.
Storage	Supabase Storage	Photos and videos for listings, blogs, and profiles.
Integrations	SendGrid (email), Google Maps (maps and	Proven third-party services.

Layer	Technology	Reason for choice
	directions), OpenWeather (dashboard weather)	
Reports	pdfkit, ExcelJS	PDF and Excel analytics and feedback exports.
Hosting	Vercel (with scheduled cron jobs) [confirm]	Serverless hosting with HTTPS.
Tools	Visual Studio Code, Git, ESLint, Node test runner	Development, version control, quality checks.

Table 24. Development Tools and Technologies

Design of Software, System, Product, or Processes

A. Conceptual Design of the Proposed System

The system has five actors and the following major modules: Places Directory, Events, Alerts, Community (reviews, blogs, forums), Feedback and Inquiries, Notifications, Moderation, Analytics, Accounts and Roles, and Settings and Security.

[Insert context diagram: Connect Daet at the center; Visitor, Registered Tourist, Tourism Officer, Moderator, Administrator; external services Supabase, Web Push service, SendGrid, Google Maps, OpenWeather]

Figure 4. Context Diagram

[Insert use case diagram of the five actors and their use cases from Table 17]

Figure 5. Proposed System Use Case

B. Logical Design of the Proposed System

The logical design describes the data flows and processes independent of the platform.

[Insert DFD showing processes: 1.0 Manage Accounts, 2.0 Manage Places and Events, 3.0 Broadcast Alerts, 4.0 Community Engagement, 5.0 Moderate Content, 6.0 Handle Inquiries, 7.0 Generate Analytics; data stores for users, listings, events, alerts, notifications, content, reports, audit log]

Figure 6. Data Flow Diagram (Level 0 and Level 1)

[Insert ERD of the core entities in Table 25]

Figure 7. Entity-Relationship Diagram

[Insert activity diagram: officer composes → chooses category, severity, audience → send now or schedule → system writes inbox rows → requests push → records counts → officer may edit, cancel, or mark all clear]

Figure 8. Activity Diagram – Sending an Alert

[Insert activity diagram: tourist submits → status pending → moderator approves or rejects → author notified → published content visible]

Figure 9. Activity Diagram – Pre-moderated Post

C. Physical Design of the Proposed System

The physical design is presented through the database design, the architecture, network layout, and screen designs in the sections below.

Software Architecture

c.1. Database Design

The database is a PostgreSQL schema managed through versioned SQL migrations. It contains 87 tables. Row-level security is enabled on the tables, and privileged operations run through server routes or database functions. The core tables are listed below, and the complete data dictionary is presented in Appendix C.

Table	**Function**
info_users, profiles	Accounts, roles (tourist, business, tourism_officer, moderator, admin), status, and public profile data. Private columns (email, phone, address) are not readable through the public API.
staff_permission_overrides, user_session_versions	Per-person permission adjustments; session revocation on role or password change.

Table	Function
info_tourist_spots, listing_categories	Places directory (attractions, accommodations, restaurants, businesses) and managed categories.
info_events, event_registrations	Events and festivals with approval workflow; attendee registrations.
alerts, alert_categories, alert_templates, alert_preferences, push_subscriptions	Alert broadcasts with schedule, status, delivery and push counts; categories; templates; subscriber preferences and devices.
info_notifications	Notification inbox for alerts, replies, approvals, and warnings.
attraction_ratings, review_helpfulness	Reviews and ratings with official office replies; helpful votes.
info_blogs, forum_threads, forum_replies, forum_categories, content_comments, content_reactions	Community content and interactions.
info_moderation, admin_user_bans, user_warnings, user_moderation_appeals	Reports, posting bans and suspensions, warnings, and appeals.
info_inquiries, info_feedback	Visitor inquiries and feedback with status and replies.

Table	Function
user_favorites	Saved items and trip plan dates and notes.
user_points, user_badges, reward_history	Gamification.
site_page_views	Consent-based, anonymized visit analytics.
info_audit_log, auth_login_failures, user_activity_log	Audit trail, lockout tracking, sign-in history.
admin_app_settings, admin_backup_settings	System settings and scheduled backups.

Table 25. Core Database Tables

c.2. Software Design Pattern

- **Client–Server with a Backend-for-Frontend API.** React pages call Next.js Route Handlers, which validate input and authorization before reading or writing the database with a server-only key.
- **Proxy / Gatekeeper.** A request proxy verifies the signed session, its version, idle timeout, page permissions, and maintenance mode before any protected page loads.
- **Policy (single rules file).** One RBAC rules module defines roles, permissions, page rules, and API rules, used by the proxy, the APIs, and the menus.

- **Publish–Subscribe (Observer).** Database changes are published through Supabase Realtime; subscribed pages refresh. Alerts fan out to subscribers through one database function and one push request.
- **State machine.** Alerts move through draft → scheduled → sent → (all clear) or cancelled; community posts through pending → published or rejected.

c.3. User Interface Design

The interface is mobile-first. It uses a blue primary color for main actions, green for success, yellow/amber for warnings, and red for errors and critical alerts, on white and light-gray surfaces with rounded cards. Buttons and tap targets are at least 44 pixels high. Public pages use a top header and a bottom navigation bar on phones (Home, Explore, Map, Profile); the administrative console uses a collapsible left sidebar. The system supports light and dark themes, reduced motion, and English, Filipino, and Bikol interface labels.

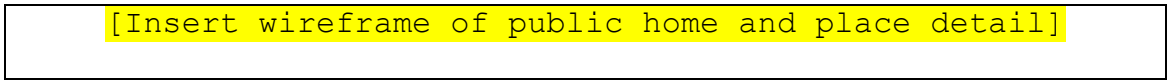


Figure 10. Layout of the Public Portal (Mobile and Desktop)

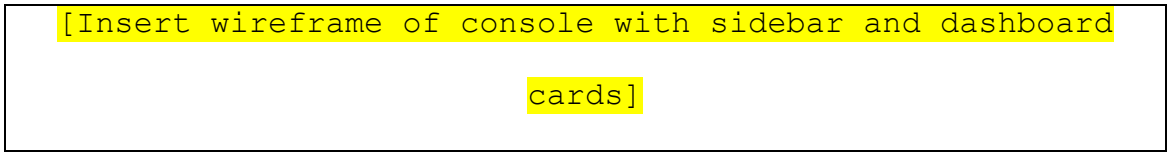


Figure 11. Layout of the Administrative Console

c.4. Architecture Design

Users' browsers connect over HTTPS to the Next.js application on the cloud host. The application serves pages and API routes; API routes connect to Supabase for the database, authentication, and storage. Browsers also hold a WebSocket connection to Supabase Realtime. When an alert is sent, a database function writes the inbox rows and calls a Supabase Edge Function secured by a shared secret, which sends push messages through the browsers' push services. Scheduled jobs (pg_cron and cloud cron) deliver scheduled alerts, run backups, and process account deletions.

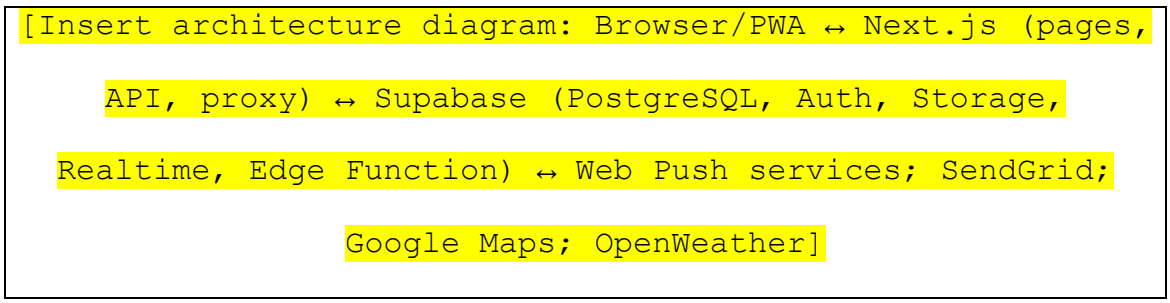


Figure 12. System Architecture

c.5. Hardware Infrastructure

No on-premise server is required. The application and database run on cloud infrastructure. The tourism office needs only computers or phones with a browser and internet access.

c.6. Network Connectivity Design

All traffic uses HTTPS (TLS). Office devices connect through the office network or mobile data to the internet; visitors connect through mobile data or Wi-Fi. Real-time updates use secure WebSockets (WSS).

[Insert network diagram: office PCs and visitor phones → router/ISP or mobile network → internet → cloud host and Supabase]

Figure 13. Network Diagram

c.7. Minimum Hardware, Software, and Network Requirements

Item	Minimum	Recommended
Computer (office)	Dual-core CPU, 4 GB RAM	Quad-core CPU, 8 GB RAM
Mobile device	Android 8 or iOS 16.4 (for push) with 2 GB RAM	Android 11+ or iOS 17+, 4 GB RAM
Browser	Current Chrome, Edge, Firefox, or Safari	Latest Chrome or Safari
Internet	5 Mbps	20 Mbps or higher
Hosting / services	Supabase and cloud host free tiers	Paid tiers for backups, bandwidth, and uptime

Table 26. Minimum and Recommended Requirements [confirm against your test devices]

Development and Testing

Development followed the seven sprints described in Chapter 3. The following verification methods were used:

- **Unit and rule tests.** 192 automated tests (Node.js test runner) cover permission rules for each role, settings validation, password policy, session idle timeout, upload content checks, error-message sanitization, alert input rules, analytics exports, and migration safety rules. All tests passed.
- **Static analysis and build.** ESLint reported no errors, and the production build compiled successfully.
- **Role access tests.** Every administrative API returned 401 without a session, and role rules deny pages and APIs outside each role's permissions.
- **Security review.** Checks based on the OWASP Top 10 led to: server-side authorization on all admin APIs; login lockout shared by sign-in, account recovery, password change, and deletion; rate limits on registration, verification emails, reports, and guest alert sign-up; restriction of notification links to the site; sanitization of blog HTML; file-signature checks on uploads; hiding of private account columns; and replacement of raw database errors with safe messages.

- **Black-box tests.** [Insert the test case table: ID, module, steps, expected result, actual result, pass/fail, for each role.]

ID	Test case	Expected result	Result
TC-01	Visitor filters restaurants by barangay	Only matching listings are shown	[ ]
TC-02	Officer schedules an alert for the weather category	Subscribers of that category receive it at the scheduled time	[ ]
TC-03	Tourist submits a forum post	Post is pending until a moderator approves it	[ ]
TC-04	Moderator suspends a user for 3 days	User is signed out and cannot sign in until expiry	[ ]
TC-05	Five wrong passwords	Account sign-in is locked for the configured duration	[ ]
TC-06	Tourist opens an admin URL	Access denied	[ ]
TC-07	Officer exports analytics to Excel	File downloads with traffic, ratings, and alert reach sheets	[ ]

Table 27. Sample Black-Box Test Cases

Description of the Prototype

[For each screen, insert a screenshot and a short description of what the user sees and can do.] Recommended screens:

- Visitor home page and places directory with tabs (See & Do, Stay, Eat & Drink, Shops & Services) and filters.
- Place detail page with photos, map, directions, hours, fees, reviews, and office replies.
- Public alerts page and an alert detail.
- Registration, sign-in, and password recovery.
- Registered tourist dashboard, saved items and trip planner, notification inbox, and feedback page.
- Blog and forum pages.
- Tourism Officer dashboard, places management, alert composer and history, reviews page, and analytics with export.
- Moderator dashboard, moderation queues, restrict dialog, and community guidelines editor.
- Administrator overview, users and permissions, settings (general, categories, security, notifications, email), failed sign-ins, audit logs, and backups.

Implementation Plan

1. **Environment setup:** create the production Supabase project and cloud host; configure environment variables (Supabase keys, session secret, SendGrid, VAPID keys, OpenWeather, cron secret).
2. **Database migration:** apply all migrations in order; deploy the push Edge Function and store its URL and secret in Supabase Vault; verify the scheduled jobs.
3. **Data migration:** encode existing attractions, establishments, and events from the office's records and Facebook page into the directory; upload photos.
4. **Accounts:** create administrator, Tourism Officer, and Moderator accounts; configure settings, categories, and community guidelines.
5. **Training:** conduct hands-on sessions for each staff role, with a user manual. [schedule]
6. **Pilot and launch:** pilot with selected users for [period], then announce the official portal through the office's Facebook page.
7. **Maintenance:** monitor audit logs and backups; apply updates.

Activity	Responsible	Target date
Environment and database setup	Developers	[ ]
Content encoding	Tourism office staff with developers	[ ]
Training	Developers	[ ]
Pilot testing	Tourism office and selected users	[ ]
Official launch	Daet Municipal Tourism Office	[ ]

Table 28. Implementation Schedule

Implementation Results

[Present the results of the pilot deployment: the period, the number of users, and measurable activity recorded by the system (e.g., listings and events published, alerts sent and their reach, inquiries answered, feedback received). Then summarize the usability evaluation using Table 23 and the open-ended comments of the tourism office personnel and users.]

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of the findings discussed in the previous chapter, the conclusions drawn, and the recommendations of the study.

Summary of Findings

SOP 1. The needs-assessment survey of 20 residents and tourists showed that tourism information about Daet was obtained mainly from Facebook (80.00%) and from friends or family (40.00%), while only 5.00% obtained it directly from the Daet Municipal Tourism Office. Seventy percent (70.00%) encountered outdated or incorrect information at least sometimes, and 80.00% had missed an important announcement about an event, road closure, or emergency. Questions were answered mostly through Google (50.00%) or friends and family (35.00%) rather than the office. Feedback was given mostly verbally or through Facebook (45.00% each), 30.00% found it hard to give feedback, and the problems most often noticed at tourist spots were cleanliness (60.00%) and poor facilities (55.00%).

SOP 2. Ninety-five percent (95.00%) of the respondents would use a central tourism platform and prefer to access it

on a mobile phone. The most useful features were the list of attractions (65.00%), photos and videos (55.00%), maps and directions (40.00%), the events calendar (35.00%), and emergency alerts (30.00%); the most needed listing information was the location (95.00%) and entrance fees (55.00%). Most respondents prefer in-app notifications for urgent announcements (65.00%) and expect updates as soon as changes happen or daily (80.00%), while 40.00% were concerned about privacy when registering. From these results, the study identified functional requirements for five user roles, technical requirements for a mobile-responsive PWA with real-time updates and push notifications, and security requirements including server-side role-based access control, row-level security, login lockout, rate limiting, input sanitization, upload checks, and audit logging.

SOP 3. The system was developed in seven Agile sprints using Next.js and Supabase, producing a places and events portal, a real-time alert module with scheduling, targeting, and reach tracking, pre-moderated community features, and a role-based administrative console with analytics exports. It passed all 192 automated tests, compiled without errors, and denied unauthorized access to all administrative APIs.

SOP 4. [State the overall weighted mean of Usability and of each of the three areas (Understandability and

Learnability; Operability and Real-Time Interaction; Interface, Accessibility, and Error Recovery) with their verbal interpretations, and the highest- and lowest-rated statements, based on the actual questionnaire results of the 42 respondents.]

Conclusions

SOP 1. Based on the findings, it is concluded that the dependence on social media and word of mouth for tourism information in Daet results in outdated information, missed advisories, inquiries that do not reach the tourism office, and feedback that is not recorded or used for decisions.

SOP 2. It is concluded that a real-time tourism system for the municipality must be mobile-first, must present complete and officially maintained listing information, must deliver alerts through in-app notifications, and must combine role-specific functions with layered security and privacy protection enforced on the server and in the database, not only in the user interface.

SOP 3. It is concluded that a single locally managed web platform can centralize tourism information, deliver targeted real-time alerts, and support moderated engagement and data-driven management.

SOP 4. [Write a categorical, number-free conclusion based on the actual results, e.g., whether the developed system is usable for its intended users and which area needs the most improvement.]

Recommendations

1. The **Daet Municipal Tourism Office** should adopt Connect Daet as its official tourism portal, assign staff to keep listings, events, and alerts current, and promote the portal through its Facebook page and printed materials such as QR codes at attractions.
2. The **Daet Municipal Tourism Office** should post links to the portal's listings, events, and alerts on its Facebook page, since 80.00% of the respondents look for information there, so that social media leads people to one official and updated source.
3. The **Daet Municipal Tourism Office** should review the feedback and ratings collected by the system regularly and share the recurring concerns, such as cleanliness and the condition of comfort rooms and waiting areas, with the offices and establishments concerned.
4. The **system administrator** should publish a clear privacy notice on the registration page that explains what

personal data is collected and why, since 40.00% of the respondents were concerned about privacy.

5. The **LGU of Daet** should fund paid cloud hosting and database plans to ensure backups, bandwidth, and uptime, and register an official domain name.
6. The **system administrator** must apply all database migrations, keep secrets and keys rotated, review audit logs and failed sign-ins regularly, and clear legacy credential data from older records.
7. The **Moderators** should review the community guidelines and the profanity word list periodically with the tourism office.
8. The **Municipal Disaster Risk Reduction and Management Office** may coordinate with the tourism office so that urgent advisories are issued through the alert module alongside official emergency channels.
9. **Future researchers** may add SMS delivery for alerts, which 30.00% of the respondents preferred, to reach users without smartphones or mobile data; provide a simplified mode for users who are not comfortable with social media or technology, publish native mobile applications, add online booking and payment for accredited establishments, provide automatic translation of user content, extend coverage to all

municipalities of Camarines Norte, and measure the system's long-term effect on tourist arrivals and satisfaction.

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APPENDICES

APPENDIX A

INTERVIEW GUIDE AND TRANSCRIPTS

[Insert interview guide and transcripts – rename this appendix if your school requires different contents.]

APPENDIX B

USABILITY SURVEY QUESTIONNAIRE (ISO/IEC 25010:2023)

SYSTEM EVALUATION QUESTIONNAIRE

DAET TOURIST INFORMATION, ENGAGEMENT, AND MANAGEMENT SYSTEM WITH REAL-TIME UPDATES

Good day!

We are 4th-year Information Technology students conducting the evaluation of our capstone project entitled "Daet Tourist Information Engagement Management System with Real-Time Updates."

This questionnaire is designed to evaluate the usability of the developed system based on the Usability characteristic of the ISO/IEC 25010 Software Quality Model. It aims to assess how easily users can understand, learn, navigate, and interact with the system, as well as how effectively the system supports users in accessing tourist information and engaging with its features.

Your participation will help us determine the system's ease of use, clarity, and overall user experience. Your feedback will also help us identify areas that may need improvement to better serve the needs of the intended users.

Please explore and use the available features of the system before answering the questionnaire. Kindly rate each

statement based on your actual experience while using the system.

Thank you for your time, cooperation, and valuable feedback.

Usability Evaluation Questionnaire

Basis: ISO/IEC 25010 Software Quality Model — Usability

Evaluation Scale

Please place a check mark (✓) under the response that best reflects your experience.

Rating	Response
5	Strongly Agree (SA)
4	Agree (A)
3	Neutral (N)
2	Disagree (D)
1	Strongly Disagree (SDA)

Part 1: Respondent Information (Optional)

Name: _____

Role: ☐ Tourist/Visitor ☐ Local Resident ☐ Content Creator ☐ Administrator ☐ Other: _____

Date: _____

Part 2: Usability Evaluation

A. Understandability and Learnability

	Statement	5 (SA)	4 (A)	3 (N)	2 (D)	1 (SDA)
1	The system's purpose and available features are clear to me.	☐	☐	☐	☐	☐

	Statement	5 (SA)	4 (A)	3 (N)	2 (D)	1 (SDA)
2	I can easily find tourist information, such as destinations, events, and announcements.	☐	☐	☐	☐	☐
3	The menus, labels, and instructions make it easy to understand how to use the system.	☐	☐	☐	☐	☐
4	I was able to learn how to use the system without needing extensive assistance.	☐	☐	☐	☐	☐

B. Operability and Real-Time Interaction

	Statement	5 (SA)	4 (A)	3 (N)	2 (D)	1 (SDA)
1	I can complete the tasks I need—such as searching for information or engaging with content—without unnecessary steps.	☐	☐	☐	☐	☐
2	The system's controls and navigation behave consistently across its pages.	☐	☐	☐	☐	☐
3	Real-time updates and notifications are easy to notice and understand.	☐	☐	☐	☐	☐

	Statement	5 (SA)	4 (A)	3 (N)	2 (D)	1 (SDA)
4	Forms and interactive features provide clear guidance about what information is needed.	☐	☐	☐	☐	☐

C. Interface, Accessibility, and Error Recovery

	Statement	5 (SA)	4 (A)	3 (N)	2 (D)	1 (SDA)
1	The text, colors, layout, and buttons make the system comfortable to use.	☐	☐	☐	☐	☐
2	The system is easy to use on the device I used, including a mobile phone, tablet, or computer.	☐	☐	☐	☐	☐
3	When I make an error, the system explains the problem and helps me correct it.	☐	☐	☐	☐	☐

Part 3: Open-Ended Feedback

1. Which parts of the system were easiest or most difficult for you to use?

_____

_____

2. Were any instructions, controls, or real-time updates unclear? Please describe.

_____

_____

3. What changes would make the system easier or more comfortable to use?

_____

_____

THANK YOU FOR YOUR TIME AND VALUABLE FEEDBACK!

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APPENDIX C

DATA DICTIONARY

The database is a PostgreSQL schema hosted on Supabase. It contains 87 tables, listed below in alphabetical order. The list was generated from the live database schema. In the Key column, PK means primary key and FK means foreign key, followed by the referenced table and column. The Null column shows whether a value may be left empty.

Table: admin_app_settings — System settings grouped by section.

Field	Type	Key	Null	Description
id	text	PK	No	Unique identifier of the record
value	jsonb		No	Value
updated_by	uuid	FK → info_users.id	Yes	User who last updated the record
updated_at	timestamptz		No	Date and time the record was last updated

Table: admin_backup_runs — History of backup runs.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
trigger_type	text		No	Trigger type
scheduled_for	date		Yes	Scheduled for
status	text		No	Current status of the record
storage_path	text		Yes	Storage path
media_object_count	integer		No	Number of media object (default: 0)
total_bytes	bigint		No	Total bytes (default: 0)
error_message	text		Yes	Error message
started_at	timestamptz		No	Date and time of started
completed_at	timestamptz		Yes	Date and time of completed
created_by	uuid	FK → info_users.id	Yes	User who created the record

Table: admin_backup_settings — Scheduled backup settings.

Field	Type	Key	Null	Description
id	smallint	PK	No	Unique identifier of the record (default: 1)
enabled	boolean		No	Enabled (default: true)
retention_days	integer		No	Retention days (default: 7)
include_media	boolean		No	Include media (default: true)
updated_by	uuid	FK → info_users.id	Yes	User who last updated the record
updated_at	timestamptz		No	Date and time the record was last updated

Table: admin_comment_mutes — Users muted from commenting.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
reason	text		No	Reason given
muted_by	uuid	FK → info_users.id	No	Muted by
muted_at	timestamptz		No	Date and time of muted
expires_at	timestamptz		Yes	Date and time of expires
lifted_by	uuid	FK → info_users.id	Yes	Lifted by
lifted_at	timestamptz		Yes	Date and time of lifted

Table: admin_messages — Messages sent by administrators to users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
sender_user_id	uuid	FK → info_users.id	No	Related sender user
recipient_user_id	uuid	FK → info_users.id	No	Related recipient user
subject	text		No	Subject
category	text		No	Category (default: Announcement)
body	text		No	Body text
scheduled_for	timestamptz		Yes	Scheduled for
status	text		No	Current status of the record (default: delivered)
delivery_error	text		Yes	Delivery error
sent_at	timestamptz		Yes	Date and time of sent
read_at	timestamptz		Yes	Date and time of read
created_at	timestamptz		No	Date and time the record was created

Field	Type	Key	Null	Description
updated_at	timestamptz		No	Date and time the record was last updated

Table: admin_user_bans — Posting bans and suspensions of users.

Field	Type	Key	Null	Description
user_id	uuid	PK	No	User who owns the record
reason	text		No	Reason given
banned_by	uuid	FK → info_users.id	No	Banned by
banned_at	timestamptz		No	Date and time of banned
lifted_by	uuid	FK → info_users.id	Yes	Lifted by
lifted_at	timestamptz		Yes	Date and time of lifted
expires_at	timestamptz		Yes	Date and time of expires
ban_type	text		No	Ban type (default: posting)

Table: alert_categories — Categories of alerts (e.g., weather, traffic, emergency).

Field	Type	Key	Null	Description
slug	text	PK	No	Short unique text identifier used in links
name	text		No	Name
description	text		No	Description
icon	text		No	Icon (default: bell)
sort_order	integer		No	Display order (default: 100)
is_active	boolean		No	Whether the record is active (default: true)
created_at	timestamptz		No	Date and time the record was created

Table: alert_preferences — Alert categories that each subscriber wants to receive.

Field	Type	Key	Null	Description
user_id	uuid	PK	No	User who owns the record
categories	text[]		No	Categories
updated_at	timestamptz		No	Date and time the record was last updated

Table: alert_templates — Reusable message templates for alerts.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record

Field	Type	Key	Null	Description
name	text		No	Name
category	text	FK → alert_categories.slug	No	Category
severity	text		No	Severity level (default: advisory)
title	text		No	Title
message	text		No	Message text
affected_area	text		No	Affected area
created_by	uuid	FK → info_users.id	Yes	User who created the record
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: alerts — Real-time alert broadcasts with schedule, targeting, status, and delivery counts.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
title	text		No	Title
message	text		No	Message text
category	text	FK → alert_categories.slug	No	Category
severity	text		No	Severity level (default: advisory)
affected_area	text		No	Affected area
barangays	text[]		No	Barangays
audience	text		No	Audience (default: category)
status	text		No	Current status of the record (default: draft)
scheduled_for	timestamptz		Yes	Scheduled for
sent_at	timestamptz		Yes	Date and time of sent
expires_at	timestamptz		Yes	Date and time of expires
cancelled_at	timestamptz		Yes	Date and time of cancelled
cancel_reason	text		Yes	Cancel reason
all_clear_at	timestamptz		Yes	Date and time of all clear
all_clear_message	text		Yes	All clear message
edited_after_send	boolean		No	Edited after send (default: false)
template_id	uuid	FK → alert_templates.id	Yes	Related template

Field	Type	Key	Null	Description
recipients_inbox	integer		No	Recipients inbox (default: 0)
push_attempted	integer		No	Push attempted (default: 0)
push_sent	integer		No	Push sent (default: 0)
push_failed	integer		No	Push failed (default: 0)
push_removed	integer		No	Push removed (default: 0)
guest_push_sent	integer		No	Guest push sent (default: 0)
push_completed_at	timestamptz		Yes	Date and time of push completed
created_by	uuid	FK → info_users.id	Yes	User who created the record
updated_by	uuid	FK → info_users.id	Yes	User who last updated the record
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: attraction_ratings — Visitor reviews and ratings of listings, with official replies.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
attraction_type	text		No	Attraction type
attraction_id	uuid		No	Related attraction
rating	integer		No	Rating from 1 to 5
pros	text[]		Yes	Pros
cons	text[]		Yes	Cons
review_text	text		Yes	Review text
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
office_reply	text		Yes	Office reply
office_reply_by	uuid	FK → info_users.id	Yes	Office reply by
office_reply_at	timestamptz		Yes	Date and time of office reply
photo_url	text		Yes	URL of the photo

Table: auth_login_failures — Failed sign-in attempts used for account lockout.

Field	Type	Key	Null	Description
email_hash	text	PK	No	Email hash

Field	Type	Key	Null	Description
failure_count	integer		No	Number of failure
window_started_at	timestamptz		No	Date and time of window started
locked_until	timestamptz		Yes	Locked until
updated_at	timestamptz		No	Date and time the record was last updated
email_hint	text		Yes	Email hint
last_failed_at	timestamptz		Yes	Date and time of last failed

Table: blog_moderation_history — History of moderation decisions on blog posts.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
blog_id	uuid	FK → info_blogs.id	Yes	Related blog
blog_title	text		No	Blog title
blog_author_id	uuid	FK → info_users.id	Yes	Related blog author
reviewed_by	uuid	FK → info_users.id	Yes	Reviewed by
action	text		No	Action
reason	text		Yes	Reason given
created_at	timestamptz		No	Date and time the record was created
previous_status	text		Yes	Previous status
new_status	text		Yes	New status

Table: comment_point_rewards — Points awarded for comments.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
content_type	text		No	Content type
content_id	uuid		No	Related content
normalized_body	text		No	Normalized body
points	numeric		No	Points (default: 0.01)
created_at	timestamptz		No	Date and time the record was created

Table: content_comments — Comments on blogs and other content.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record

Field	Type	Key	Null	Description
content_type	text		No	Content type
content_id	uuid		No	Related content
user_id	uuid	FK → info_users.id	No	User who owns the record
parent_id	uuid	FK → content_comments.id	Yes	Related parent
body	text		No	Body text
gif_url	text		Yes	URL of the gif
sticker_url	text		Yes	URL of the sticker
is_pinned	boolean		No	Whether the record is pinned (default: false)
is_best_answer	boolean		No	Whether the record is best answer (default: false)
relevance_score	integer		No	Relevance score (default: 0)
status	text		No	Current status of the record (default: active)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
mention_data	jsonb		No	Mention data

Table: content_hashtags — Links between content and hashtags.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
hashtag_id	uuid	FK → hashtags.id	No	Related hashtag
content_type	text		No	Content type
content_id	uuid		No	Related content
created_at	timestamptz		No	Date and time the record was created

Table: content_polls — Polls attached to content.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
content_type	text		No	Content type
content_id	uuid		No	Related content
created_by	uuid	FK → info_users.id	No	User who created the record
question	text		No	Question
poll_type	text		No	Poll type (default: multiple_choice)
options	jsonb		Yes	Options

Field	Type	Key	Null	Description
correct_option_index	integer		Yes	Correct option index
results_visibility	text		No	Results visibility (default: after_vote)
duration_hours	integer		No	Duration hours (default: 168)
starts_at	timestamptz		No	Date and time of starts
ends_at	timestamptz		Yes	Date and time of ends
total_votes	integer		No	Total votes (default: 0)
is_active	boolean		No	Whether the record is active (default: true)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: content_reactions — Reactions on content.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
content_type	text		No	Content type
content_id	uuid		No	Related content
reaction_type	text		No	Reaction type
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: content_shares — Shares of content.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
content_type	text		No	Content type
content_id	uuid		No	Related content
share_type	text		No	Share type (default: repost)
caption	text		Yes	Caption
recipient_user_id	uuid	FK → info_users.id	Yes	Related recipient user
platform	text		Yes	Platform
created_at	timestamptz		No	Date and time the record was created

Table: daily_feedback_questions — Daily feedback questions shown to users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
question	text		No	Question
feedback_date	date		No	Date of feedback
is_active	boolean		No	Whether the record is active (default: true)
created_by	uuid	FK → info_users.id	Yes	User who created the record
created_at	timestamptz		No	Date and time the record was created

Table: daily_feedback_votes — Answers of users to daily feedback questions.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
question_id	uuid	FK → daily_feedback_questions.id	No	Related question
user_id	uuid	FK → info_users.id	No	User who owns the record
response	text		No	Response
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: direct_message_reactions — Reactions on direct messages.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
message_id	uuid	FK → direct_messages.id	No	Related message
user_id	uuid	FK → info_users.id	No	User who owns the record
reaction	text		No	Reaction
created_at	timestamptz		No	Date and time the record was created

Table: direct_messages — Direct messages between users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
sender_id	uuid	FK → info_users.id	No	Related sender
recipient_id	uuid	FK → info_users.id	No	Related recipient
body	text		Yes	Body text

Field	Type	Key	Null	Description
created_at	timestamptz		No	Date and time the record was created
read_at	timestamptz		Yes	Date and time of read
media_url	text		Yes	URL of the media
media_type	text		Yes	Media type
reply_to_message_id	uuid	FK → direct_messages.id	Yes	Related reply to message
metadata	jsonb		No	Additional data in JSON format
deleted_for	uuid[]		No	Deleted for
edit_locked_at	timestamptz		Yes	Date and time of edit locked

Table: event_moderation_history — History of approval decisions on events.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
event_id	uuid	FK → info_events.id	Yes	Related event
event_title	text		No	Event title
event_author_id	uuid	FK → info_users.id	Yes	Related event author
reviewed_by	uuid	FK → info_users.id	Yes	Reviewed by
action	text		No	Action
reason	text		Yes	Reason given
created_at	timestamptz		No	Date and time the record was created

Table: event_registrations — Registrations of users for events.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
event_id	uuid	FK → info_events.id	No	Related event
user_id	uuid	FK → info_users.id	No	User who owns the record
status	text		No	Current status of the record (default: confirmed)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: event_reminder_deliveries — Event reminders sent to registered users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
event_id	uuid	FK → info_events.id	No	Related event
registration_id	uuid	FK → event_registrations.id	No	Related registration
recipient_user_id	uuid	FK → info_users.id	No	Related recipient user
sent_by	uuid	FK → info_users.id	Yes	Sent by
status	text		No	Current status of the record
error_message	text		Yes	Error message
sent_at	timestamptz		Yes	Date and time of sent
created_at	timestamptz		No	Date and time the record was created

Table: follow_requests — Pending requests to follow private accounts.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
follower_id	uuid	FK → info_users.id	No	Related follower
following_id	uuid	FK → info_users.id	No	Related following
status	text		No	Current status of the record (default: pending)
created_at	timestamptz		No	Date and time the record was created
responded_at	timestamptz		Yes	Date and time of responded

Table: follows — Follow relationships between users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
follower_id	uuid	FK → info_users.id	No	Related follower
following_id	uuid	FK → info_users.id	No	Related following
created_at	timestamptz		No	Date and time the record was created

Table: forum_categories — Forum categories.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
name	text		No	Name
description	text		Yes	Description

Field	Type	Key	Null	Description
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: forum_replies — Replies to forum threads.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
thread_id	uuid	FK → forum_threads.id	No	Related thread
user_id	uuid	FK → info_users.id	No	User who owns the record
content	text		No	Content text
status	text		No	Current status of the record (default: active)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
image_url	text		Yes	URL of the image
video_url	text		Yes	URL of the video
gif_url	text		Yes	URL of the gif
sticker_url	text		Yes	URL of the sticker
mention_data	jsonb		No	Mention data
parent_reply_id	uuid	FK → forum_replies.id	Yes	Related parent reply

Table: forum_subscriptions — Users subscribed to forum threads.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
thread_id	uuid	FK → forum_threads.id	No	Related thread
user_id	uuid	FK → info_users.id	No	User who owns the record
created_at	timestamptz		No	Date and time the record was created

Table: forum_threads — Forum threads.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
title	text		No	Title
content	text		No	Content text
category_id	uuid	FK → forum_categories.id	Yes	Related category

Field	Type	Key	Null	Description
created_by	uuid	FK → info_users.id	No	User who created the record
status	text		No	Current status of the record (default: published)
reply_count	integer		No	Number of reply (default: 0)
last_activity_at	timestamptz		No	Date and time of last activity
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
views	integer		No	Views (default: 0)
is_pinned	boolean		No	Whether the record is pinned (default: false)
tags	text[]		No	Tags
image_urls	text[]		No	Image urls
location	text		Yes	Location
pinned	boolean		No	Pinned (default: false)

Table: hashtag_follows — Hashtags followed by users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
hashtag_id	uuid	FK → hashtags.id	No	Related hashtag
created_at	timestamptz		No	Date and time the record was created

Table: hashtags — Hashtags used in content.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
name	text		No	Name
usage_count	integer		No	Number of usage (default: 0)
category	text		Yes	Category
is_trending	boolean		No	Whether the record is trending (default: false)
is_branded	boolean		No	Whether the record is branded (default: false)
first_seen_at	timestamptz		No	Date and time of first seen
last_used_at	timestamptz		No	Date and time of last used

Table: info_announcements — Announcements published by staff.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
title	text		No	Title
content	text		No	Content text
announcement_type	text		No	Announcement type (default: info)
audience	text		No	Audience (default: all)
priority	integer		No	Priority (default: 1)
image_url	text		Yes	URL of the image
status	text		No	Current status of the record (default: draft)
published_at	timestamptz		Yes	Date and time of published
expires_at	timestamptz		Yes	Date and time of expires
created_by	uuid	FK → info_users.id	Yes	User who created the record
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
severity	text		No	Severity level (default: info)
scheduled_for	timestamptz		Yes	Scheduled for
video_url	text		Yes	URL of the video
view_count	integer		No	Number of view (default: 0)
open_count	integer		No	Number of open (default: 0)
engagement	integer		No	Engagement (default: 0)
tourist_spot_id	uuid	FK → info_tourist_spots .id	Yes	Optional attraction associated with a visitor-facing tourism announcement.

Table: info_audit_log — Audit trail of administrative and staff actions.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	Yes	User who owns the record
action	text		No	Action
table_name	text		No	Table name
record_id	uuid		Yes	Related record
changes	jsonb		Yes	Changes

Field	Type	Key	Null	Description
ip_address	text		Yes	Ip address
user_agent	text		Yes	User agent
created_at	timestamptz		No	Date and time the record was created

Table: info_blogs — Blog posts with moderation status.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
title	text		No	Title
slug	text		Yes	Short unique text identifier used in links
content	text		No	Content text
excerpt	text		Yes	Excerpt
featured_image	text		Yes	Featured image
category	text		No	Category (default: news)
tags	text[]		Yes	Tags
status	text		No	Current status of the record (default: draft)
published_at	timestamptz		Yes	Date and time of published
views	integer		No	Views (default: 0)
likes	integer		No	Likes (default: 0)
comments_count	integer		No	Number of comments (default: 0)
created_by	uuid	FK → info_users.id	Yes	User who created the record
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
featured	boolean		No	Featured (default: false)
images	text[]		No	Images
videos	text[]		No	Videos
media_layout	text		No	Media layout (default: swipe)
location	text		Yes	Location
approval_status	text		No	Approval status (default: approved)
submitted_at	timestamptz		Yes	Date and time of submitted
reviewed_at	timestamptz		Yes	Date and time of reviewed
reviewed_by	uuid	FK → info_users.id	Yes	Reviewed by
rejection_reason	text		Yes	Rejection reason

Table: info_comments — Comments on community posts.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
blog_id	uuid	FK → info_blogs.id	No	Related blog
user_id	uuid	FK → info_users.id	No	User who owns the record
content	text		No	Content text
status	text		No	Current status of the record (default: approved)
likes	integer		No	Likes (default: 0)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
parent_id	uuid	FK → info_comments.id	Yes	Related parent
mention_data	jsonb		No	Mention data

Table: info_events — Events and festivals with approval status.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
title	text		No	Title
description	text		Yes	Description
location	text		Yes	Location
venue	text		Yes	Venue
latitude	numeric		Yes	Latitude of the location
longitude	numeric		Yes	Longitude of the location
start_date	date		Yes	Date of start
end_date	date		Yes	Date of end
start_time	time		Yes	Start time
end_time	time		Yes	End time
category	text		No	Category (default: festival)
organizer	text		Yes	Organizer
is_free	boolean		No	Whether the record is free (default: true)
ticket_price	numeric		Yes	Ticket price (default: 0)
max_attendees	integer		Yes	Max attendees
current_attendees	integer		No	Current attendees (default: 0)
featured_image	text		Yes	Featured image

Field	Type	Key	Null	Description
images	text[]		Yes	Images
videos	text[]		Yes	Videos
status	text		No	Current status of the record (default: draft)
published_at	timestamptz		Yes	Date and time of published
created_by	uuid	FK → info_users.id	Yes	User who created the record
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
recurrence	text		Yes	Recurrence
tags	text[]		Yes	Tags
featured	boolean		No	Featured (default: false)
view_count	integer		No	Number of view (default: 0)
share_count	integer		No	Number of share (default: 0)
approval_status	text		No	Approval status (default: approved)
submitted_at	timestamptz		Yes	Date and time of submitted
reviewed_at	timestamptz		Yes	Date and time of reviewed
reviewed_by	uuid	FK → info_users.id	Yes	Reviewed by
rejection_reason	text		Yes	Rejection reason

Table: info_faqs — Frequently asked questions managed by staff.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
question	varchar		No	Question
answer	text		No	Answer
sort_order	integer		No	Display order (default: 0)
is_published	boolean		No	Whether the record is published (default: false)
created_by	uuid	FK → info_users.id	Yes	User who created the record
updated_by	uuid	FK → info_users.id	Yes	User who last updated the record
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: info_feedback — Visitor feedback with status and replies.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	Yes	User who owns the record
title	text		Yes	Title
message	text		No	Message text
category	text		No	Category (default: general)
rating	integer		Yes	Rating from 1 to 5
status	text		No	Current status of the record (default: open)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
tourist_spot_id	uuid	FK → info_tourist_spots .id	Yes	Optional active destination associated with public tourism feedback.
admin_response	text		Yes	Private response recorded by tourism-office reviewers.
responded_by	uuid	FK → info_users.id	Yes	Responded by
responded_at	timestamptz		Yes	Date and time of responded
image_urls	text[]		No	Image urls
location	text		Yes	Location

Table: info_inquiries — Visitor inquiries with status and replies.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
title	text		Yes	Title
message	text		No	Message text
category	text		No	Category (default: general)
status	text		No	Current status of the record (default: open)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
admin_response	text		Yes	Admin response
responded_at	timestamptz		Yes	Date and time of responded
responded_by	uuid	FK → info_users.id	Yes	Responded by

Field	Type	Key	Null	Description
image_urls	text[]		No	Image urls

Table: info_moderation — Content and account reports and their resolution.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
report_type	text		No	Report type (default: content)
reason	text		No	Reason given
description	text		Yes	Description
severity	text		No	Severity level (default: medium)
reported_by	uuid	FK → info_users.id	Yes	Reported by
reported_user_id	uuid	FK → info_users.id	Yes	Related reported user
reported_item_id	uuid		Yes	Related reported item
reported_item_table	text		Yes	Reported item table
status	text		No	Current status of the record (default: pending)
assigned_to	uuid	FK → info_users.id	Yes	Assigned to
moderation_notes	text		Yes	Moderation notes
resolution	text		Yes	Resolution
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
resolved_at	timestamptz		Yes	Date and time of resolved

Table: info_notifications — Notification inbox entries for alerts, replies, approvals, and warnings.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
title	text		No	Title
message	text		No	Message text
type	text		No	Type of the record (default: info)
is_read	boolean		No	Whether the record is read (default: false)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
link	text		Yes	Link

Field	Type	Key	Null	Description
post_id	uuid		Yes	Related post
actor_id	uuid	FK → info_users.id	Yes	Related actor
comment_id	uuid	FK → content_comments.id	Yes	Related comment
reply_id	uuid	FK → content_comments.id	Yes	Related reply
post_owner_id	uuid		Yes	Related post owner
admin_message_id	uuid	FK → admin_messages.id	Yes	Related admin message
alert_id	uuid	FK → alerts.id	Yes	Related alert

Table: info_password_resets — Password reset requests.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
token	text		No	Token
email	text		No	E-mail address (private)
expires_at	timestamptz		No	Date and time of expires
used	boolean		No	Used (default: false)
used_at	timestamptz		Yes	Date and time of used
created_at	timestamptz		No	Date and time the record was created

Table: info_tourist_spots — Places directory: attractions, accommodations, restaurants, and businesses.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
name	text		No	Name
slug	text		Yes	Short unique text identifier used in links
description	text		Yes	Description
category	text		No	Category (default: beach)
location	text		Yes	Location
latitude	numeric		Yes	Latitude of the location
longitude	numeric		Yes	Longitude of the location
entry_fee	numeric		Yes	Entry fee (default: 0)
opening_hours	text		Yes	Opening hours
best_visit_time	text		Yes	Best visit time
accessibility_info	text		Yes	Accessibility info

Field	Type	Key	Null	Description
featured_image	text		Yes	Featured image
images	text[]		Yes	Images
videos	text[]		Yes	Videos
rating	numeric		Yes	Rating from 1 to 5 (default: 0)
review_count	integer		No	Number of review (default: 0)
visit_count	integer		No	Number of visit (default: 0)
status	text		No	Current status of the record (default: active)
featured	boolean		No	Featured (default: false)
created_by	uuid	FK → info_users.id	Yes	User who created the record
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
gallery_images	text[]		Yes	Gallery images
image_url	text		Yes	URL of the image
contact_number	text		Yes	Contact number
save_count	integer		No	Number of save (default: 0)
share_count	integer		No	Number of share (default: 0)
activities	text		Yes	Administrator-provided activities available at this attraction.
transportation_info	text		Yes	Administrator-provided transportation guidance for this attraction.
entry_fee_verified	boolean		No	Whether the displayed entry_fee was confirmed by an administrator. (default: false)
listing_type	text		No	Listing type (default: attraction)
tags	text[]		No	Tags
barangay	text		Yes	Barangay where the place or event is located
email	text		Yes	E-mail address (private)
website	text		Yes	Website
facebook_url	text		Yes	URL of the facebook
price_range	text		Yes	Price range

Table: info_user_posts — Community posts of users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
title	text		No	Title
content	text		No	Content text
status	text		No	Current status of the record (default: published)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
visibility	text		No	Visibility (default: public)

Table: info_users — User accounts with role, status, and profile details. Private columns are not readable through the public API.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
email	text		No	E-mail address (private)
password	text		Yes	Legacy password field; sign-in uses Supabase Auth (to be cleared)
full_name	text		No	Full name
user_type	text		No	User type (default: tourist)
status	text		No	Current status of the record (default: active)
profile_image_url	text		Yes	URL of the profile image
phone_number	text		Yes	Phone number (private)
bio	text		Yes	Bio
address	text		Yes	Address
city	text		Yes	City
country	text		Yes	Country
points	numeric		No	Points (default: 0)
last_login	timestamptz		Yes	Last login
is_online	boolean		No	Whether the record is online (default: false)
email_verified	boolean		No	Email verified (default: false)
created_at	timestamptz		No	Date and time the record was created

Field	Type	Key	Null	Description
updated_at	timestamptz		No	Date and time the record was last updated
level	integer		No	Level (default: 1)
onboarding_completed	boolean		No	Onboarding completed (default: false)
account_deactivated_at	timestamptz		Yes	Set when the account owner deactivates their account.
deletion_requested_at	timestamptz		Yes	When the account owner requested account deletion.
deletion_scheduled_for	timestamptz		Yes	Account deletion deadline: 30 days after an owner request or 15 days after an administrator schedules deletion.
deletion_reminder_7_claimed_at	timestamptz		Yes	Date and time of deletion reminder 7 claimed
deletion_reminder_7_sent_at	timestamptz		Yes	Date and time of deletion reminder 7 sent
deletion_reminder_1_claimed_at	timestamptz		Yes	Date and time of deletion reminder 1 claimed
deletion_reminder_1_sent_at	timestamptz		Yes	Date and time of deletion reminder 1 sent
deletion_processing_at	timestamptz		Yes	Date and time of deletion processing
username	text		Yes	Optional public username; unique without regard to letter case.

Table: listing_categories — Categories managed by staff for the places directory.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
listing_type	text		No	Listing type
slug	text		No	Short unique text identifier used in links
name	text		No	Name
is_active	boolean		No	Whether the record is active (default: true)
sort_order	integer		No	Display order (default: 100)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: mentions — User mentions in content.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
mentioned_user_id	uuid	FK → info_users.id	No	Related mentioned user

Field	Type	Key	Null	Description
mentioned_by_user_id	uuid	FK → info_users.id	No	Related mentioned by user
content_type	text		No	Content type
content_id	uuid		No	Related content
mention_text	text		Yes	Mention text
is_read	boolean		No	Whether the record is read (default: false)
created_at	timestamptz		No	Date and time the record was created

Table: message_conversation_settings — Per-conversation settings such as mute.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
other_user_id	uuid	FK → info_users.id	No	Related other user
is_archived	boolean		No	Whether the record is archived (default: false)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
is_muted	boolean		No	Whether the record is muted (default: false)
is_deleted	boolean		No	Whether the record is deleted (default: false)

Table: notifications — General user notifications.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid		No	User who owns the record
title	text		No	Title (default: CONNECT-Daet)
body	text		No	Body text
url	text		No	URL (default: /)
payload	jsonb		No	Payload
created_at	timestamptz		No	Date and time the record was created

Table: password_reset_rate_limits — Rate-limit counters for password reset requests.

Field	Type	Key	Null	Description
bucket_key	text	PK	No	Bucket key
window_started_at	timestamptz		No	Date and time of window started

Field	Type	Key	Null	Description
request_count	integer		No	Number of request
updated_at	timestamptz		No	Date and time the record was last updated

Table: password_reset_token_uses — Records of used password reset tokens to prevent reuse.

Field	Type	Key	Null	Description
token_hash	text	PK	No	Token hash
consumed_at	timestamptz		No	Date and time of consumed

Table: pinned_message_activity — Activity history of pinned messages.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
conversation_id	text		No	Related conversation
message_id	uuid		No	Related message
actor_user_id	uuid	FK → info_users.id	No	Related actor user
action_type	text		No	Action type
created_at	timestamptz		No	Date and time the record was created

Table: pinned_messages — Pinned messages in conversations.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
conversation_id	text		No	Related conversation
message_id	uuid		No	Related message
pinned_by	uuid	FK → info_users.id	No	Pinned by
pinned_at	timestamptz		No	Date and time of pinned
actor_user_id	uuid	FK → info_users.id	No	Related actor user

Table: poll_votes — Votes on content polls.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
poll_id	uuid	FK → content_polls.id	No	Related poll
user_id	uuid	FK → info_users.id	No	User who owns the record
selected_option_index	integer		Yes	Selected option index
answer_text	text		Yes	Answer text
is_correct	boolean		Yes	Whether the record is correct (default: false)

Field	Type	Key	Null	Description
created_at	timestamptz		No	Date and time the record was created

Table: profiles — Public profile data linked to Supabase Auth users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
full_name	text		Yes	Full name
profile_image_url	text		Yes	URL of the profile image
bio	text		Yes	Bio
city	text		Yes	City
country	text		Yes	Country
address	text		Yes	Address
is_public	boolean		No	Whether the record is public (default: true)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
cover_photo_url	text		Yes	URL of the cover photo
location	text		Yes	Location
privacy_level	text		No	Privacy level (default: public)
language_preference	text		No	Language preference (default: en)
notification_preferences	jsonb		No	Notification preferences
privacy_preferences	jsonb		No	User settings > Privacy: { privacyLevel, showOnlineStatus, showActivity, allowMessagesFrom, allowMentions, searchable }
ui_preferences	jsonb		No	User settings > Appearance: { theme: light|dark|system, reduceMotion: boolean }

Table: push_subscriptions — Browser push subscriptions of devices that receive alerts.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid		Yes	User who owns the record
subscription_endpoint	text		No	Subscription endpoint

Field	Type	Key	Null	Description
subscription	jsonb		No	Subscription
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
p256dh	text		Yes	Public key of the push subscription
auth	text		Yes	Authentication secret of the push subscription
alert_categories	text[]		Yes	Alert categories
last_seen_at	timestamptz		No	Date and time of last seen

Table: reposts — Reposts of community posts.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
original_content_type	text		No	Original content type
original_content_id	uuid		No	Related original content
quote_text	text		Yes	Quote text
created_at	timestamptz		No	Date and time the record was created
status	text		No	Current status of the record (default: active)
visibility	text		No	Visibility (default: public)

Table: review_helpfulness — Helpful votes on reviews.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
review_type	text		No	Review type
review_id	uuid		No	Related review
user_id	uuid	FK → info_users.id	No	User who owns the record
is_helpful	boolean		No	Whether the record is helpful (default: true)
created_at	timestamptz		No	Date and time the record was created

Table: reward_history — History of points earned and spent.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
subsystem_source	text		No	Subsystem source (default: system)

Field	Type	Key	Null	Description
points_earned	numeric		No	Points earned (default: 0)
description	text		Yes	Description
created_at	timestamptz		No	Date and time the record was created

Table: reward_point_awards — Points awarded for specific actions.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
action_key	text		No	Action key
entity_type	text		No	Entity type
entity_id	uuid		No	Related entity
points	numeric		No	Points
created_at	timestamptz		No	Date and time the record was created

Table: reward_redemptions — Rewards redeemed by users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
reward_id	text		No	Related reward
reward_title	text		No	Reward title
points_spent	integer		No	Points spent
created_at	timestamptz		No	Date and time the record was created

Table: share_analytics — Counts of shares by channel for analytics.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
content_type	text		No	Content type
content_id	uuid		No	Related content
platform	text		No	Platform
share_count	integer		No	Number of share (default: 0)
last_shared_at	timestamptz		Yes	Date and time of last shared

Table: site_maintenance_settings — Maintenance mode settings.

Field	Type	Key	Null	Description
id	smallint	PK	No	Unique identifier of the record (default: 1)
enabled	boolean		No	Enabled (default: false)
message	text		No	Message text
updated_by	uuid	FK → info_users.id	Yes	User who last updated the record
updated_at	timestamptz		No	Date and time the record was last updated

Table: site_page_views — Consent-based, anonymized page-view analytics.

Field	Type	Key	Null	Description
visitor_hash	text		No	Visitor hash
page_path	text		No	Page path
viewed_at	timestamptz		No	Date and time of viewed
device_type	text		No	Device type (default: unknown)

Table: staff_chat_messages — Internal chat messages between staff.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
sender_id	uuid	FK → info_users.id	No	Related sender
body	text		No	Body text
created_at	timestamptz		No	Date and time the record was created

Table: staff_permission_overrides — Per-person permission adjustments granted to or removed from staff accounts.

Field	Type	Key	Null	Description
user_id	uuid	PK	No	User who owns the record
permission	text	PK	No	Permission
granted	boolean		No	Granted
updated_by	uuid	FK → info_users.id	Yes	User who last updated the record
updated_at	timestamptz		No	Date and time the record was last updated

Table: system_categories — General categories used by system modules.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
name	text		No	Name
icon_emoji	text		Yes	Icon emoji
sort_order	integer		No	Display order (default: 0)
is_active	boolean		No	Whether the record is active (default: true)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated

Table: tourist_spot_account_views — Views of listings by signed-in accounts, used for analytics.

Field	Type	Key	Null	Description
tourist_spot_id	uuid	PK	No	Related tourist spot
account_hash	text	PK	No	Account hash
first_viewed_at	timestamptz		No	Date and time of first viewed
last_viewed_at	timestamptz		No	Date and time of last viewed

Table: tourist_spot_geocode_cache — Cached map coordinates of listing addresses.

Field	Type	Key	Null	Description
query_key	text	PK	No	Query key
latitude	float8		No	Latitude of the location
longitude	float8		No	Longitude of the location
expires_at	timestamptz		No	Date and time of expires
cached_at	timestamptz		No	Date and time of cached

Table: tourist_spot_geocode_rate_limit — Rate-limit counters for address geocoding requests.

Field	Type	Key	Null	Description
id	smallint	PK	No	Unique identifier of the record
next_allowed_at	timestamptz		No	Date and time of next allowed

Table: user_activity_log — Sign-in and account activity history of users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
activity_type	text		No	Activity type
entity_type	text		Yes	Entity type
entity_id	uuid		Yes	Related entity
description	text		Yes	Description
metadata	jsonb		No	Additional data in JSON format
created_at	timestamptz		No	Date and time the record was created

Table: user_badges — Badges earned by users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
badge_name	text		No	Badge name
badge_icon	text		Yes	Badge icon
description	text		Yes	Description
earned_at	timestamptz		No	Date and time of earned

Table: user_blocks — Users blocked by other users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
blocker_id	uuid	FK → info_users.id	No	Related blocker
blocked_id	uuid	FK → info_users.id	No	Related blocked
created_at	timestamptz		No	Date and time the record was created

Table: user_favorites — Saved places and events, with trip plan dates and notes.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
item_type	text		No	Item type
item_id	uuid		No	Related item
created_at	timestamptz		No	Date and time the record was created
planned_visit_date	date		Yes	Date of planned visit

Field	Type	Key	Null	Description
trip_notes	text		Yes	Trip notes

Table: user_feed_preferences — Feed settings of each user.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
preferred_categories	text[]		No	Preferred categories
enabled	boolean		No	Enabled (default: true)
created_at	timestamptz		No	Date and time the record was created
updated_at	timestamptz		No	Date and time the record was last updated
favorite_places	text[]		No	Favorite places
onboarding_completed	boolean		No	Onboarding completed (default: false)

Table: user_follows — Follow relationships between users (community feed).

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
follower_id	uuid	FK → info_users.id	No	Related follower
following_id	uuid	FK → info_users.id	No	Related following
created_at	timestamptz		No	Date and time the record was created

Table: user_moderation_appeals — Appeals of users against moderation actions.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
reason	text		No	Reason given
status	text		No	Current status of the record (default: pending)
decision_notes	text		Yes	Decision notes
reviewed_by	uuid	FK → info_users.id	Yes	Reviewed by
reviewed_at	timestamptz		Yes	Date and time of reviewed
created_at	timestamptz		No	Date and time the record was created

Table: user_points — Points earned by users (gamification).

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record

Field	Type	Key	Null	Description
user_id	uuid	FK → info_users.id	No	User who owns the record
points	numeric		No	Points (default: 0)
reason	text		Yes	Reason given
source	text		Yes	Source
created_at	timestamptz		No	Date and time the record was created

Table: user_poll_votes — Votes on user polls.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
poll_id	uuid	FK → user_polls.id	No	Related poll
user_id	uuid	FK → info_users.id	No	User who owns the record
selected_option_index	integer		No	Selected option index
created_at	timestamptz		No	Date and time the record was created

Table: user_polls — Polls created by users.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
created_by	uuid	FK → info_users.id	No	User who created the record
question	text		No	Question
options	text[]		No	Options
category	text		Yes	Category
is_closed	boolean		No	Whether the record is closed (default: false)
total_votes	integer		No	Total votes (default: 0)
created_at	timestamptz		No	Date and time the record was created
ends_at	timestamptz		Yes	Date and time of ends

Table: user_session_versions — Session version per user, used to sign out users after a role, password, or status change.

Field	Type	Key	Null	Description
user_id	uuid	PK	No	User who owns the record
version	integer		No	Version (default: 0)
updated_at	timestamptz		No	Date and time the record was last updated

Table: user_storage_objects — Files uploaded by users to storage.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
bucket_id	text		No	Related bucket
object_path	text		No	Object path
created_at	timestamptz		No	Date and time the record was created
file_size	bigint		Yes	File size
content_type	text		Yes	Content type

Table: user_warnings — Warnings issued to users by moderators.

Field	Type	Key	Null	Description
id	uuid	PK	No	Unique identifier of the record
user_id	uuid	FK → info_users.id	No	User who owns the record
reason	text		No	Reason given
report_id	uuid		Yes	Related report
issued_by	uuid	FK → info_users.id	Yes	Issued by
created_at	timestamptz		No	Date and time the record was created

Table: visitor_profiles — Additional profile data of visitors and tourists.

Field	Type	Key	Null	Description
user_id	uuid	PK	No	User who owns the record
age_range	text		Yes	Age range
residence_area	text		Yes	Residence area
consent_given	boolean		No	Consent given (default: false)
consented_at	timestamptz		No	Date and time of consented
updated_at	timestamptz		No	Date and time the record was last updated

APPENDIX D

USER MANUAL

[Insert user manual – rename this appendix if your school requires different contents.]

CURRICULUM VITAE

[Insert each proponent's curriculum vitae.]